



Atellica® Solution LAS Interface Guide

Atellica Solution

LAS Interface Guide

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1 Introduction

This guide provides information on connecting an Atellica® Solution fitted with Atellica Sample Handler Connect (SHC) to a Laboratory Automation System (LAS).

An LAS is any external, automated sample transport system that is connected to an Atellica Solution. The combined Atellica Solution with SHC is ready to integrate with an LAS.

An SHC is an Atellica Solution module that provides a physical and electrical communication connection to the LAS. The SHC is the transfer interface for sample containers onto the Atellica Solution via the Atellica Magline® Transport. The SHC module also provides sample container characterization.

NOTE: The SHC module does not perform any sample testing or provide results.

For additional information, refer to the Atellica Solution Operator's Guide or online help.

2 Capabilities

Capabilities represent key functionality that modules, subsystems, and components provide. The capabilities are categorized into LAS and LIS capabilities.

About LAS Capabilities

The LAS capabilities on the Atellica Solution are the following:

LAS communication management	Notes
Listen for Communication from the LAS	When the LAS is not operational, the Atellica Solution enters a listening mode while waiting to handshake with the LAS (Page 82 <i>About Establishing LAS Connection</i>).
Display LAS Communication Status	The Atellica Solution displays the communication status with the LAS. The possible modes are: • Listening • Non-remote control • Remote control Non-remote control: • The LAS establishes a connection to receive information. • The LAS does not present or receive samples. • This mode displays on the Atellica Solution or the LAS. • The operator must stop using the LAS connection for transferring samples, but must send the LAS information about the Atellica Solution. Remote control mode: • The LAS establishes connection and receives information. • The system actively transfers samples in and out of the Atellica Solution.

LAS communication management	Notes
Initialize LAS Communication Handshaking	The Atellica Solution and the LAS initialize the communication channels prior to sharing information and coordinating work (Page 82 <i>About Establishing LAS Connection</i>) (Page 87 <i>About Initializing the LAS Connection</i>).
Reset LAS Communication	If communication errors occur, the LAS and the Atellica Solution re-initialize communication between them as part of error recovery (Page 107 <i>About LAS Connection Reset</i>).
Atellica Solution Sets Remote Control Mode	The Atellica Solution has 2 operational modes with respect to the LAS. While in Non-Remote Control mode, the Atellica Solution transmits system health information to the LAS, but it is not able to process samples from the LAS. While in Remote Control mode, the Atellica Solution transmits system health information to the LAS and it is able to process samples from the LAS. The Atellica Solution sets its LAS mode of operation (Page 96 <i>About Remote Control Status</i>).
LAS sample management	Notes
Load Samples from the LAS	The action of transferring samples to the Atellica Solution from the LAS (Page 120 <i>Instrument Control</i>).
Unload Samples to the LAS	The action of transferring samples from the Atellica Solution to the LAS (Page 120 <i>Instrument Control</i>).
System information management for LAS	Notes
Send Assay Availability to the LAS	In addition to documenting inventory, the assay is available if it is onboard and the QC is acceptable. Status is red, green, or yellow based on availability (Page 100 <i>About Test Inventory Messages</i>).

System information management for LAS		Notes
Send System Status to the LAS		System Status indicates if the Atellica Solution is operational. A legacy LAS may use this information in its routing logic. Eliminating this information requires changes on those systems (Page 89 <i>Instrument Health Message</i>).
	Send LAS Interfacing Status	Atellica Solution sends the LAS the status of the SHC that moves samples from the LAS to the Atellica Magline Transport (Page 89 <i>Instrument Health Message</i>).
	Send Instrument Status	Not Mandatory: • The Atellica Solution sends the LAS a specific status where the Atellica Solution rejects new work to cause off-loading of samples. • Can accommodate operation when the Atellica Solution moves samples on the SH . (Page 89 <i>Instrument Health Message</i>)
	Send Remote Control Status	The Atellica Solution sends the LAS the current Remote Control Status. An operator can set this status on the Atellica Solution (Page 89 <i>Instrument Health Message</i>).
	Onboard Specimen Inventory Status	Specimen Inventory Status is a count of how many samples are onboard the Atellica Solution from the LAS.
	Listen for Communication from the LAS	When LAS is not operational, the Atellica Solution enters a listening mode while waiting to handshake with the LAS (Page 82 <i>About Establishing LAS Connection</i>).

About LIS Capabilities

The following is a list of capabilities supported with the LIS/Middleware interface. Capabilities that apply to an Atellica Solution are noted.

LIS/Middleware interface capabilities	Notes
Synchronize Time Servers for Daylight Saving Time	Synchronizes clocks across multiple computer systems (LIS, LAS, Middleware, and Atellica Solution). The Atellica Solution (PCC) and an Atellica Data Manager (ADM) server perform synchronizing.
Allow Operator Searches	The operator can perform middleware database searches from the Middleware screen by Sample ID, Physician Name, Patient ID, and so on. NOTE: The Atellica DM supports such searches on the Atellica Solution.
Operator Configures Feature Access Levels	The laboratory manager can limit access of certain middleware information and functionality to specific operators. These access levels are internal to Middleware.
Manage Test Results	
Operator Review Results	Reviewable using the Atellica DM.
Release Test Results	Any released results are reported as described in SMN 11312743, <i>Atellica Solution LIS Interface Guide</i> .
Manage Rerun/Reflex Test Orders	Managing does not interact with the LIS Interface. The Atellica Solution receives orders as described in SMN 11312743, <i>Atellica Solution LIS Interface Guide</i> .
Autoverification of Test Results	After receipt of results from the Atellica Solution, the Middleware solution automatically releases results to the clinician using various criteria. The Atellica Solution supports Auto-verification. The Atellica DM supports Auto-verification and Auto-validation.

LIS/Middleware interface capabilities	Notes
Verify Reference Interval	Check returned result against reference interval for each assay in the database. Supported by the Atellica Solution.
Perform Delta Checks on Results	Middleware checks the returned result against known results of the same assay for the same patient ID. Middleware highlights results that are more than a preset threshold different for human inspection. Not a capability of all middleware solutions. Supported by the Atellica Solution.
Verify Instrument Flags	Middleware checks for any system flags on results before releasing to the clinician. Not a capability of all middleware solutions. Supported by the Atellica Solution.
Verify QC Severity	Patient result is held based on the QC result associated with that test. The patient results are not a capability of all middleware solutions. Supported by the Atellica Solution.
Customizable Verification Checks	The Atellica DM performs this operation. Checks performed with information in the Middleware database. Information in the Middleware database can come from various sources.
Manage Patient Information	Patient Information is communicated as described in SMN 11312743, Atellica Solution <i>LIS Interface Guide</i>.

LIS/Middleware interface capabilities	Notes
Operator Enter Patient Info	—
Operator Edit Patient Info	—
Receive Patient Information from the LIS	—
Filter Patient Information Flow to Atellica Solution	To protect patient privacy, middleware removes patient-specific information from the data stream that is transmitted to the Atellica Solution.
Manage QC Information	Control Order information is described in SMN 11312743, <i>Atellica Solution LIS Interface Guide</i> .
Associate QC with Test Result	Option that enables middleware to record the QC results associated with a particular test result for a patient. Not a capability of all middleware solutions.
View QC Information	Supported in the Atellica Solution.
Send QC Information to External Databases	Middleware can send QC results and reports to an external entity. The Atellica Solution transmits QC results to the QC material supplier for tracking purposes. Verify if the laboratory's middleware supports this feature. This is further described in SMN 11312743, <i>Atellica Solution LIS Interface Guide</i>
Customize Rule Sets for QC Checking	Laboratory personnel can customize how the middleware solution checks if QC is acceptable. For example, the Atellica DM can use the result of the QC result of one assay to support or reject the QC of one or more other tests (QC Blocking).

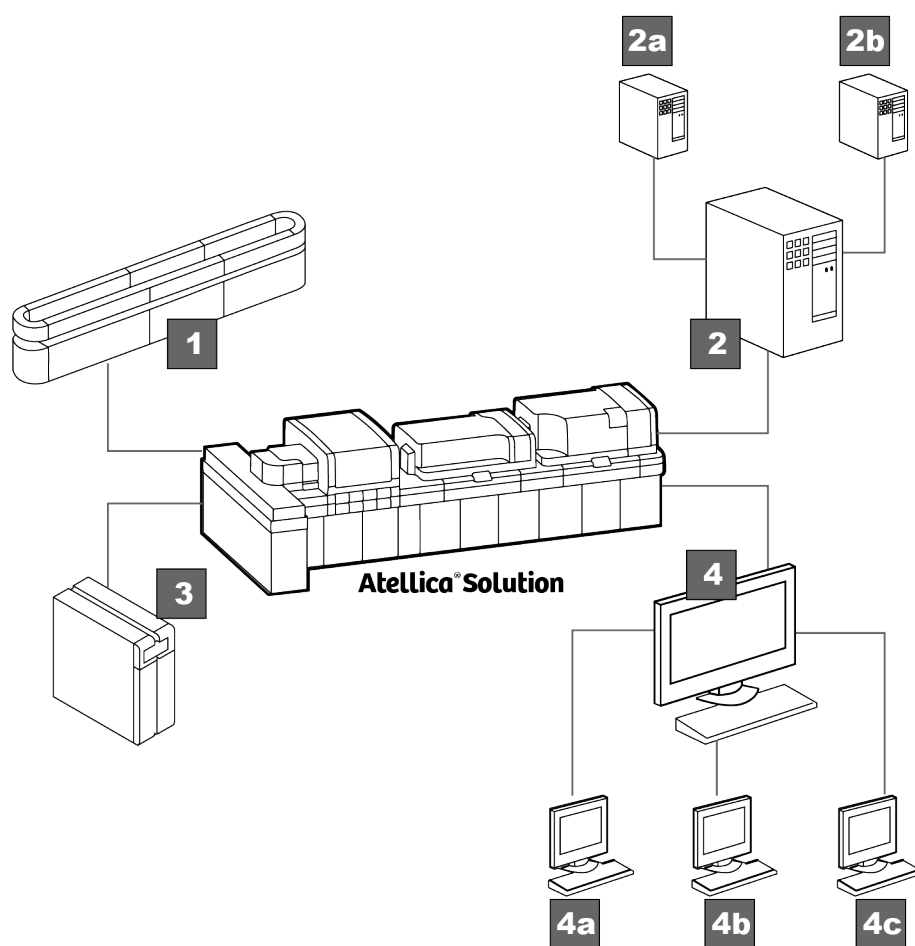
LIS/Middleware interface capabilities	Notes
Enable or Disable Tests Based on QC Results	—
Automation Sample Management Responsibilities with the Atellica Solution	
Monitors Status of Atellica Solution Attached to Automation	—
Monitors Status of Samples as They Are Transported By LAS	—
Determine Relative Routing Logic for Samples on Siemens Healthineers Automation	—
Provide Test Orders to LAS	—
Order New Tests or Actions Based on Sample Status or Error Flags	Atellica DM can take action based on status of samples and errors the Atellica Solution reports.
Manage Test Orders	
Operator-entered Test Orders	Test orders are described in SMN 11312743, Atellica Solution LIS Interface Guide.
Receive Test Orders from the LIS	Test orders are described in SMN 11312743, Atellica Solution LIS Interface Guide.
Transmit Test Orders to Atellica Solution	Test orders are described in SMN 11312743, Atellica Solution LIS Interface Guide.
Transmit Test Orders to LAS	Test orders are described in SMN 11312743, Atellica Solution LIS Interface Guide.

LIS/Middleware interface capabilities	Notes
Target Test Orders to a Specific Atellica Solution	Middleware solution can direct an order to a specific Atellica Solution attached to automation. For the Atellica Solution, the middleware solution can also direct an order to a specific module within an Atellica Solution. The Atellica Solution uses field O25 in Test Order Record as described in SMN 11312743, Atellica Solution <i>LIS Interface Guide</i>.
Manage Pre- and Post-Processing Tests	Based on the Atellica Solution internal architecture, pre- and post-analytic processing (for example, decapping or recapping) shares the same workflow as performing tests. As such, if a middleware solution is required to control these activities, the operators must present them as test orders to the Atellica Solution.

3 Atellica Solution Interface Specifications

Atellica Solution Context

Context diagram for LAS and LIS/Middleware connections to an Atellica Solution.



-
- 1 LAS
 - 2 LIS
 - 2a. LIS-HL7
 - 2b. LIS-ASTM

- 3 Lab Inventory Manager
 - 4 Middleware
 - 4a. 3rd Party Middleware
 - 4b. Atellica DM
 - 4c. Data Management System
-

About Interface Entities

The entities for consideration within the scope of this document are the LAS and LIS.

The general constraints of the LAS are:

- If an Atellica Solution is connected to an LAS, the Atellica Solution must use a middleware solution.
- The Atellica Solution does not support connectivity to ADVIA® Automation and StreamLAB® Automation.

The general constraints of the LIS are:

- An Atellica Solution may not have an LIS and Middleware directly connected at the same time.

The Atellica Solution only supports LIS-HL7- and LIS-ASTM-compliant LIS types.

The global constraint for the Middleware connection is:

- The Atellica Solution cannot connect to an LIS and Middleware at the same time.

The connecting Middleware are:

- Atellica Data Manager
- Inpeco Data Management System (or also called Nemo 2)

NOTE: When an LAS is in use, it uses the Atellica DM or the DMS middleware.

About LAS Top Level Interface

The LAS top level interface has the following characteristics:

- Interface type: External.
- Direction: Bi-directional.
- Constraint: The Atellica Solution is a pick-and-place connection. It is not point-in-space sampling.

About Mechanical Interface

The mechanical LAS interface is structured into the following items:

- LAS interface: Mechanical force applied
- LAS interface: Mechanical interface - Positions
- LAS Interface: Personnel access

About Mechanical Interface: Positions and Tube Bottom Location (Automation)

- Reference point: The Atellica Solution is at a variable position with respect to the sample container and aligned to the LAS. Do not align the LAS to the Atellica Solution.
- DeviceA: Atellica Solution.
- DeviceB: Center of tube on automation track.
- Min Z position: 825 mm for Tube bottom.
- Max Z position: 895 mm for Tube bottom.

- Position description:
 - AngularUnits: String = degrees
 - Device A: String = Atellica Solution
 - Device B: String = Center of tube
 - Max X position: Float
 - Max Y position: Float
 - Max Z position: Float = 895 mm
 - MaxTotalAngularError: Float
 - MaxTotalPositionError: Float
 - MinX position: Float
 - MinYposition: Float
 - MinZ position: Float = 815 mm
 - PositionUnits: String = mm
 - Reference point: String = Floor at Front
- Constraints:
 - Mechanism lifts tube 110 mm above the bottom position.
 - The Atellica Solution covers allow a 110-mm tube to move through the interface point unobstructed.

Electrical Interface: Communication – RS232

- Protocol = RS-232
- ConnectorType = Female DB9

About LAS Software Interface

The LAS Software Interface is as follows:

- CommunicationProtocol: Universal Router Analyzer Protocol (uRAP) protocol
- Software communication: For missing values, refer to the LAS Interface Software section (Page 73 *LAS Interface Software*).
 - CommunicationProtocol: String = uRAP protocol
 - DataTransferBlockMaxSize: Int
 - DataTransferSizeUnits: String
 - DataTransferRate: float
 - DataTransferUnits: String

NOTE: The Atellica Solution creates and maintains a log file of all LAS communication messages to and from the LAS with appropriate “human readable” annotation for a time period of 30 days.

About LIS Interface

LIS top level interface has the following characteristics:

- Interface type: External
- Direction: Bi-directional
- Structural interface:
 - Category
 - Context: InterfaceType = external
 - Direction: Bi-directional
- Constraints:
 - None

About Electrical Interface

Electrical Interface: Communication - Ethernet

- Protocol = Ethernet
- Communication:
 - CableType : String
 - ConnectorType: String
 - Number of wires: Int
 - Protocol: String = Ethernet
 - WireGage: Int
 - BaudRate: 57600 Baud

Electrical Interface: Communication – RS232

- Protocol = RS-232 via a terminal concentrator
- ConnectorType = Female DB99
- Communication:
 - CableType : String
 - ConnectorType: String = Female DB9
 - Number of wires: Int
 - Protocol: String = RS-232
 - WireGage: Int

LIS/Middleware SW Interface Options

The LIS/Middleware interface is structured into the following items:

LIS/Middleware software interface: ASTM protocol

CommunicationProtocol = ASTM

ASTM1381	127-character data frames
LIS01-A2	64,000-character data frames

SW communication

CommunicationProtocol	String = ASTM
DataTransferBlockMaxSize	Int
DataTransferSizeUnits	String

DataTransferRate	float
DataTransferUnits	String
LIS/Middleware software interface: HL7 protocol	
CommunicationProtocol = HL7	
SW communication	
CommunicationProtocol	String = HL7
DataTransferBlockMaxSize	Int
DataTransferSizeUnits	String
DataTransferRate	float
DataTransferUnits	String

About Supported Sample Containers



CAUTION

Do not use unapproved sample containers. The use of unapproved sample containers may result in damage to the system. Use only approved sample containers.

The system supports the following types of sample containers:

- Primary tubes
- Cal-QC tubes
- Pour-off tubes
- Tube-top sample cups
- Capillary collection containers
- False-bottom tubes

NOTE: If the system does not include a Decapper, capped sample containers are not supported. The operator must remove caps from sample containers before loading the containers in racks. The system does not support Decappers in configurations connected to Aptio.

Primary Tubes

The system supports the following primary tubes:

NOTE: Container type measurements are in mm.

13x75 Vacutainer Hemogard - BD 367983

Material	Plastic
Cap type	Safety
Volume	3.5 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a teal 13x75 rack insert.
ESL	Yes
Decapper	Yes

13x75 Vacutainer - BD 367981

Material	Plastic
Cap type	Stopper
Volume	3.5 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a teal 13x75 rack insert.
ESL	Yes
Decapper	No

13x75 Vacuette - Greiner 454067

Material	Plastic
Cap type	Safety
Volume	4 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a teal 13x75 rack insert.
ESL	Yes
Decapper	Yes

13x100 Vacutainer Hemogard - BD 367977

Material	Plastic
Cap type	Safety
Volume	6 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a brown 13x100 rack insert.
ESL	Yes
Decapper	Yes

13x100 Vacutainer - BD 368660

Material	Plastic
Cap type	Stopper
Volume	6 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a brown 13x100 rack insert.
ESL	Yes
Decapper	No

13x100 Vacuette - Greiner 456073

Material	Plastic
Cap type	Safety
Volume	5 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a brown 13x100 rack insert.
ESL	Yes
Decapper	Yes

16x100 Vacutainer Hemogard - BD 362799

Material	Plastic
Cap type	Safety
Volume	8.5 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes
ESL	Yes
Decapper	Yes

16x100 Vacutainer glass - BD 366430

Material	Glass
Cap type	Stopper
Volume	10 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes
ESL	Yes
Decapper	No

16x100 Vacutainer - BD 367820

Material	Plastic
Cap type	Stopper
Volume	10 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes
ESL	Yes
Decapper	No

16x100 Vacuette - Greiner 455010

Material	Plastic
Cap type	Safety
Volume	9 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes
ESL	Yes
Decapper	Yes

16x92 S-Monovette - Sarstedt 02.1388.100

Material	Plastic
Cap type	Thread
Volume	9 mL
Bottom shape	Syringe
SH	Yes
SHC	Yes
DL	No
ESL	Yes
Decapper	No

13x90 S-Monovette - Sarstedt 04.1935.001

Material	Plastic
Cap type	Thread
Volume	4.9 mL
Bottom shape	Syringe
SH	Yes
SHC	Yes
DL	Yes Requires a brown 13x100 rack insert.
ESL	Yes
Decapper	Yes

15x75 S-Monovette - Sarstedt 03.1397.100

Material	Plastic
Cap type	Thread
Volume	5.5 mL
Bottom shape	Syringe
SH	Yes
SHC	Yes
DL	No
ESL	Yes
Decapper	Yes

15x92 S-Monovette - Sarstedt 01.1602.001

Material	Plastic
Cap type	Thread
Volume	7.5 mL
Bottom shape	Syringe
SH	Yes
SHC	Yes
DL	Yes
ESL	Yes
Decapper	Yes

Cal-QC Tubes

The system supports the following Cal-QC tubes:

NOTE: To maintain the stability of Atellica calibrator or QC materials, use a glass sample container for onboard storage.

NOTE: Container type measurements are in mm.

12x75 glass Cal-QC tube for onboard storage

Material	Glass
Cap type	None
Volume	5 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a grey 12x75 rack insert.
ESL	Yes
Decapper	No

Bio-Rad control tube

Material	Plastic
Cap type	Thread
Volume	4 mL
Bottom shape	Round
SH	Yes
SHC	No
DL	Yes Requires a grey 12x75 rack insert.
ESL	Yes
Decapper	Yes

Pour-Off Tubes

The system supports the following pour-off tubes:

NOTE: Container type measurements are in mm.

12x75 Plastic: VWR 82050-128, Greiner 115101

Material	Plastic
Cap type	None
Volume	5 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a grey 12x75 rack insert.
ESL	Yes
Decapper	No

12x75 Glass: VWR 60826-836, Corning 99445-12

Material	Glass
Cap type	None
Volume	6 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a grey 12x75 rack insert.
ESL	Yes
Decapper	No

13x100 Plastic: VWR 82050-202, Greiner 125101

Material	Plastic
Cap type	None
Volume	9 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a brown 13x100 rack insert.
ESL	Yes
Decapper	No

13x100 Glass: VWR 60826-847, Corning 99445-13

Material	Glass
Cap type	None
Volume	9 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes Requires a brown 13x100 rack insert.
ESL	Yes
Decapper	No

14x100 Plastic: VWR 82050-228, Greiner 136101

Material	Plastic
Cap type	None
Volume	5 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	No
ESL	Yes
Decapper	No

16x75 Plastic: VWR 89206-934, Globe 119040A

Material	Plastic
Cap type	None
Volume	8 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	No
ESL	Yes
Decapper	No

16x100 Plastic: VWR 82050-234, Greiner 160101

Material	Plastic
Cap type	None
Volume	10 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes
ESL	Yes
Decapper	No

16x100 Glass: VWR 60826-869, Corning 99445-16

Material	Glass
Cap type	None
Volume	10 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	Yes
ESL	Yes
Decapper	No

15.3x92 Transfer Tube - Sarstedt 60.610.100

Material	Plastic
Cap type	Thread
Volume	10 mL
Bottom shape	Round
SH	Yes
SHC	Yes
DL	No
ESL	Yes
Decapper	No

Tube-Top Sample Cups

The system supports the following tube-top sample cups:

NOTE: Only use 1 mL tube-top sample cups with 13 mm sample containers.

Atellica Solution Tube-Top Sample Cups (1 mL) REF 11069061

Material	Plastic
Cap type	None
Volume	1 mL
Bottom shape	Conical
SH	Yes
SHC	No
DL	Yes
ESL	Yes
Decapper	No

NOTE: Only use 2 mL tube-top sample cups with 16 mm sample containers.

Atellica Solution Tube-Top Sample Cups (2 mL) REF 11069062

Material	Plastic
Cap type	None
Volume	2 mL
Bottom shape	Conical
SH	Yes
SHC	No
DL	Yes
ESL	Yes
Decapper	No

Capillary Collection Containers

The system supports the following capillary collection containers:

NOTE: The system must include at least 1 IM Analyzer to support the use of capillary collection containers.

NOTE: The operator must place capillary collection containers in an appropriate holder before loading them in an SH rack.

- For false-bottom capillary collection containers, use the green Holder Tube Capillary False Bottom (11069063).
- For plain conical-bottom capillary collection containers, use the blue Holder Tube Capillary (11069064).

Becton Dickinson Microtainer 365967

Material	Plastic
Cap type	Stopper
Volume	1 mL
Bottom shape	Conical
Required holder	False-bottom capillary tube holder (green)
SH	Yes
SHC	No
DL	No
ESL	No
Decapper	No

Terumo Capiject T-MG

Material	Plastic
Cap type	Stopper
Volume	1 mL
Bottom shape	Conical
Required holder	Capillary tube holder (blue)
SH	Yes
SHC	No
DL	No
ESL	No
Decapper	No

Greiner MiniCollect 450472

Material	Plastic
Cap type	Stopper
Volume	1 mL
Bottom shape	Conical
Required holder	Capillary tube holder (blue)
SH	Yes
SHC	No
DL	No
ESL	No
Decapper	No

NOTE: The Greiner MiniCollect Complete must load in a special false-bottom tube rack due to the construction of the sample container.

Greiner MiniCollect Complete 450550

Material	Plastic
Cap type	Stopper
Volume	0.8 mL
Bottom shape	Conical (false-bottom)
Required holder	None
	NOTE: The capillary collection container comes preassembled with a 13x75 carrier tube and does not require a capillary tube holder.
SH	Yes
SHC	No
DL	No
ESL	No
Decapper	No

Sarstedt Microvette 20.1344

Material	Plastic
Cap type	Stopper
Volume	1 mL
Bottom shape	Conical
Required holder	False-bottom capillary tube holder (green)
SH	Yes
SHC	No
DL	No
ESL	No
Decapper	No

Ram Scientific Safe-T-Fill 07 7121

Material	Plastic
Cap type	Stopper
Volume	1 mL
Bottom shape	Conical
Required holder	Capillary tube holder (blue)
SH	Yes
SHC	No
DL	No
ESL	No
Decapper	No

False-Bottom Tubes

The system supports the following false-bottom tubes:

15.3x92 False-Bottom - Sarstedt 62.612

Material	Plastic
Cap type	Screw
Volume	4 mL
Bottom shape	Flat
SH	Yes
SHC	No
DL	Yes
ESL	No
Decapper	No

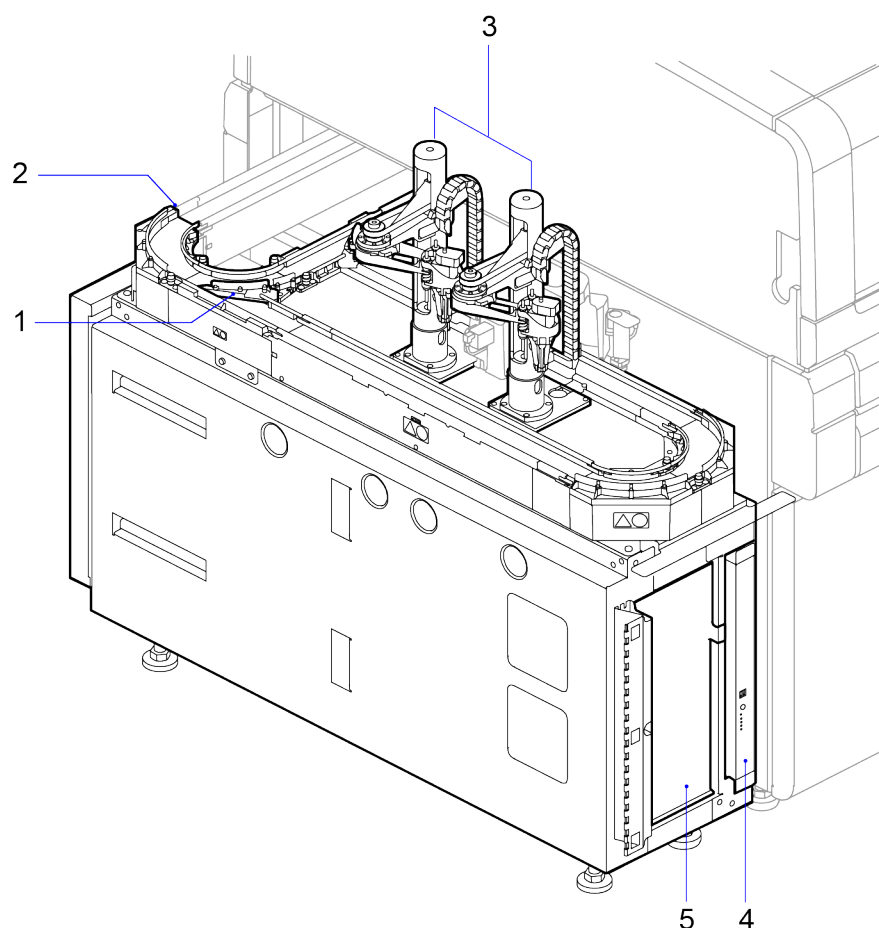
4 SHC Interface Specification

The Atellica Sample Handler Connect (SHC) module provides physical and communication connection to the Lab Automation System buffer (LAS buffer).

The SHC conducts sample container characterization and transfers sample containers to the Atellica Solution. The SHC module does not perform any sample testing or provide results.

The following figure depicts the entities interfacing with the SHC:

SHC module



- 1 Atellica Magline Transport diversion
- 2 SH connection

- 3 Folding arms
 - 4 Module Manager PC
 - 5 Controller Chassis
-

Within the Atellica Solution, the SHC interfaces with the SHC frame, Planner, System Instrument Manager (SIM), and Atellica Magline Transport. External to the Atellica Solution, the SHC interfaces with the LAS Buffer, and external power.

NOTE: The scope of this interface specification document describes the essential interfaces.

- **Atellica Magline Transport:** Other modules in the Atellica Solution deliver and receive sample containers using the Atellica Magline Transport. The SHC robot interfaces with the Atellica Magline Transport to pick and place sample containers into and out of the Atellica Magline Transport carriers from the LAS.
- **SHC Frame:** The SHC module is connected to the SHC frame to ensure alignment and stability.

SHC Theory of Operations

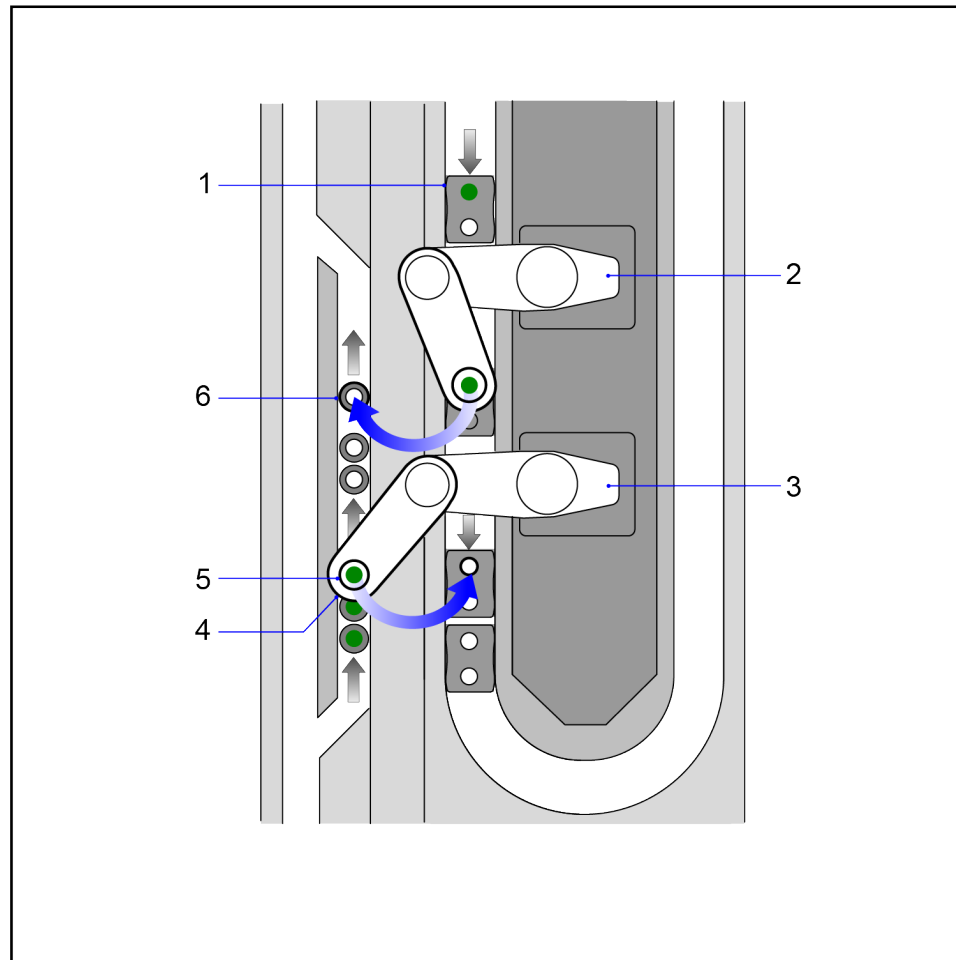
The following sections detail the SHC automation LAS connection theory of operation and related timing.

About SHC Theory of Operations

The SHC provides a transfer interface from the LAS Buffer to the Atellica Magline Transport. The sample interaction “Pick and Place” loads the sample container from the LAS Buffer onto the Atellica Magline Transport for processing and unloads the sample container to the LAS Buffer when the Atellica Solution completes sample processing.

The SHC 2-axes (Z and Theta) transfer arms with electrically actuated grippers. An additional axis is manually adjustable for arm alignment to the LAS Interface Positions. One transfer arm loads sample containers onto the Atellica Solution. The second transfer arm moves sample containers to the LAS. The LAS instrument buffer has 2 corresponding Interface Positions.

SHC LAS interface schematic Top view



1	Atellica Magline Transport	4	LAS Track buffer
2	SHC Output Transfer Arm (P1)	5	LAS Input Pick Position
3	SHC Input Transfer Arm (P0)	6	LAS Output Place Position

About Pick-and-Place Activity

All SHC pick-and-place activity is in response to messages the LAS sends. The normal operating scenario is:

1. The input transfer arm loads the sample container from the LAS buffer Interface Position (IPO).
2. The gripper confirms the tube is picked from the LAS buffer Interface Position.

3. The gripper moves the sample container to the Atellica Magline Transport.
4. The Atellica Magline Transport routes the sample container to the Tube Characterization Station (TCS) and the Atellica analyzers for processing.
5. The SHC input transfer arm continues to load sample containers onto the Atellica Magline Transport as space permits.
6. Upon completion of sample processing for sample containers received from the LAS, the Atellica Solution routes the sample container to the SHC output transfer arm position.
7. The output transfer arm unloads the sample containers to the LAS Buffer Interface Position (IP1).

NOTE: The LAS is responsible for managing the LAS buffer.

SHC Timing

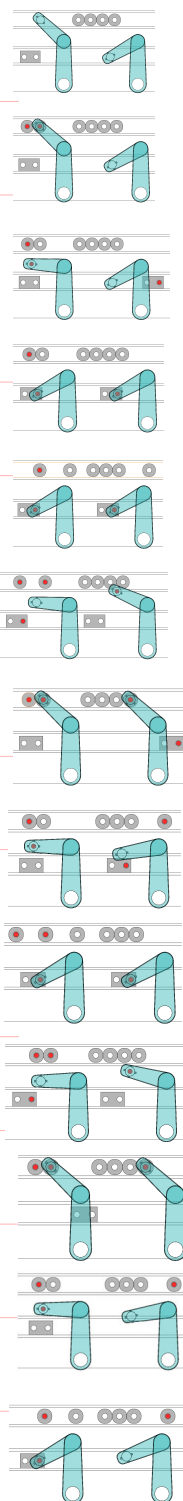
The maximum transfer rate is 1200 sample container exchanges per hour (1200 sample containers the Atellica Solution simultaneously loads and unloads to the LAS). The physical arm timing estimates for an Atellica Solution operating at maximum throughput is shown in the following table. Different analyzer states and capacities may limit the number of consecutive transfers at a maximum throughput. The transfer arm cycle times to complete a transfer in or out is 3 seconds with error conditions being the only exceptions.

Move	Description	Z-Axis			R-Axis			G-Axis			Track			Duration (s)	Time (s)	Carrier
		D	P	D	P	D	P	D	P	D	D	P	P			
		0	0	0	0	0	0	0	0	0	0	0	0	0	0	
1	Descend to tube	-70	-70	0	0	0	0	0	0	0	0	0	0	0.3	0.3	
2	Grab tube	0	-70	0	0	-10	-10	-10	-10	0	0	0	0	0.3	0.6	
3	Move up to travel plane	70	0	0	0	0	0	0	-10	0	0	0	0	0.3	0.9	
4	Rotate to carrier place position	0	0	-70	-70	0	-70	0	-10	0.2	0.2	0.2	0.2	0.4	1.3	
5	Bottom tube in carrier place position	-70	-70	0	-70	0	-70	0	-10	0.2	0.2	0.4	0.5	0.5	1.8	0.9
6	Release gripper	0	-70	0	-70	10	-70	10	0	0.2	0.2	0.6	0.3	0.3	2.1	1.2

Move	Descript ion	Z-Axis		R-Axis		G-Axis		Track		Durati on (s)	Time (s)	Carrier
		D	P	D	P	D	P	D	P			Time (s)
7	Move to clear carrier height	70	0	0	-70	0	0	0.2	0.8	0.3	2.4	1.5
8	Return to Pick Position	0	0	0	-70	0	0	0.2	1	0.4	2.8	1.9
	Cycle Time (1/2 full rotation)									2.8		
								TPH		1285.7		
D = Distance (mm)												
P = Position (degrees)												
TPH = tubes per hour												

The following are example message exchanges for the transfer scenario.
The message permits ample time for track cycling to index pallets.

LAS P0	Robot P-0	LAS P1	Robot P-1
—	—	—	Start position
—	—	—	Descend to tube on the Atellica Solution
—	—	—	Grab tube
Sends Exchange Message	—	Sends Exchange Message	Move to travel plane
Receives Exchange Message ACK	Start position	Receives Exchange Message ACK	Move to track place position
LAS locks Pallet	Descend to tube on track	LAS locks Pallet	Bottom tube in track pallets
LAS locks Pallet	Grab tube	LAS locks Pallet	Release gripper
LAS locks Pallet	Move to travel plane	LAS locks Pallet	Move to travel plane
Receives Exchange Message Response	Move to Atellica Solution place position	Receives Exchange Message Response	Return to start position
Sends Exchange Message Response ACK	Bottom tube in Atellica Solution sample carrier	Sends Exchange Message Response ACK	—
LAS locks Pallet	Release gripper	LAS locks Pallet	—
Index P-0	Move to travel plane	Index P-1	—
Read new pallets at P-0	Return to start position	Read new pallets at P-1	—
Sends Exchange Message	—	Sends Exchange Message	—



The Atellica Solution has a 3-second cycle time. When connected to an LAS, the Atellica Solution transfer mechanism transfers up to 1200 sample containers per hour. Loading SH tubes and LAS tubes simultaneously reduces the sample container transfer rate. The Atellica Solution prioritizes samples based on the following criteria:

About Priority of Sample Container Loading

1. Stat at SH
2. Stat at SHC
3. ASAP at SH (supported on Atellica Solution)
4. Routine at SHC
5. Routine at SH

NOTE: For the Atellica® CH Analyzer, configurations below the 1200 tests per hour threshold drop the LAS interface throughput relative to the Atellica Solution capacity and test usage when running concurrently with SH-loaded samples.

About Specific Timing Requirements for SHC and Automation Connections

The following are required for the SHC to achieve the tube exchange throughput:

1. For IP0 interface position, a LOAD-UNLOAD COMMAND RESPONSE message is sent by the Atellica Solution less than 2 seconds after LOAD-UNLOAD COMMAND sent by LAS.
2. For IP1 interface position, LOAD-UNLOAD COMMAND RESPONSE message sent by the Atellica Solution less than 2 seconds after LOAD-UNLOAD COMMAND sent by LAS.
3. COMMAND sent by LAS.
4. IP0 LOAD-UNLOAD COMMAND RESPONSE message sent by the Atellica Solution offset less than 3 seconds from IP1 LOAD-UNLOAD COMMAND RESPONSE message sent by the Atellica Solution. (IP0 needs to complete within 3 seconds of IP1)

Depending on the number of Atellica analyzers and the tests ordered per sample, the actual sample container load and unload rate may be less than 1200 sample containers per hour. Additionally, samples the operator loads directly on the Atellica Solution using the SH drawers have loading priority over samples the LAS delivers.

SHC Physical Connection

About Connection Structure

The connection structure of the SHC is self-contained with the exception of the following external interfaces:

- Operator input to the access door
- Utility Station communication connection for Planner
- LAS communication connection for:
 - uRAP
 - Atellica Magline Transport communication to module PC
 - AC power input to the SHC

The SH connects to the SHC using 2 power feeds:

- The SH provides AC power to the SHC AC distribution board. This is conditioned power for distribution to the low voltage power supplies.
- The SH UPS is connected to the SHC module manager PC.

NOTE: The UPS distributes shutdown commands for the Module Manager PC over Ethernet from the PCC.

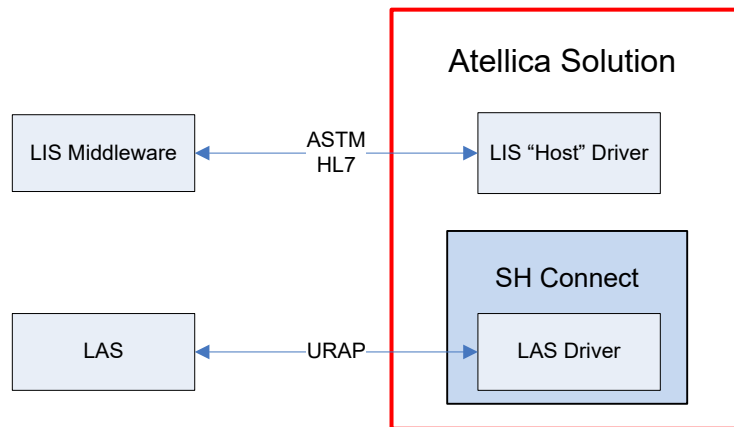
About Communication Connections

The following figure displays the simplified communication structure for the Atellica Solution with an SHC relative to an Atellica Solution and an LAS buffer.

- The Planner is the backbone of the analytical platform application server, which coordinates activities across the various module types within the Atellica Solution. It sends action requests to the Sample Source and Sinks, passes sample information, and is responsible for routing the sample tubes containers to the specific Analytical Modules for processing.
- The LAS-Driver is an application within SHC with user interface integrated in the Atellica Solution UI. It provides functionality to:
 - Configure the LAS connection parameters
 - Display the communication flow in a terminal window
 - Communicate high-level events with the Central Control and the uRAP library.

- The uRAP Library is a library for translation of high level data to uRAP records and vice versa, and controls the communication between an Atellica Solution and the LAS buffer at a low level.
- The host driver is an application in the DM layer that provides functionality to configure the parameters for the LIS connection. It supports logging of communication and communicates with the DM on high level with events. For the LIS, the system uses middleware (DM or DMS) when an Atellica Solution is connected to the LAS.

SHC Connection Architecture



About Connection to LAS

The SHC connects to the LAS buffer over an Ethernet (TCP/IP) if automation supports it or over a serial (RS-232) connection (Page 53 *Connection Pinouts*) (Page 73 *LAS Interface Software*).

About Connection to LIS

An Atellica Solution is connected to a LIS over either Ethernet or RS-232 (Page 53 *Connection Pinouts*). The LIS connection setup determines the settings of the connection to the LIS. Communication formats between an LIS and an Atellica Solution are selected as HL7 and ASTM.

Connection Pinouts

The following sections detail the physical characteristics and location of the connections.

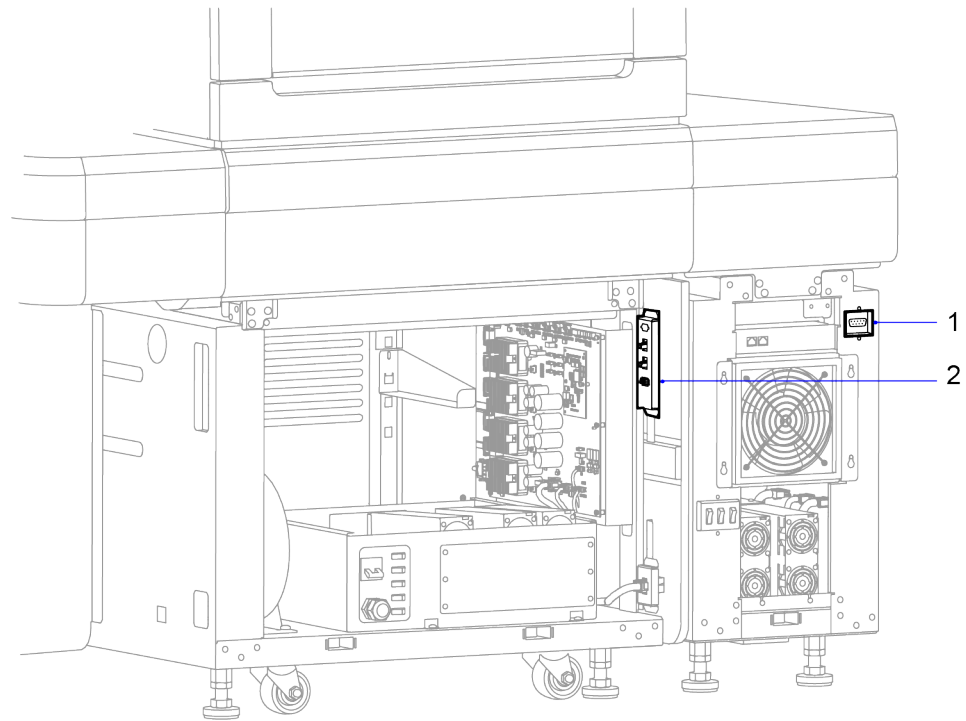
About LAS Serial Connection Pinouts

NOTE: For a connection, only RXD, TXD and GND pins are supported.

A suitable LAS connection cable is included with the Atellica Solution Interface Module. The SHC does not supply a serial cable in its kit. The Atellica Solution has a female DB9 and Ethernet interface on the back panel of the module. The location of the connector is shown on the back of the system in the figure below.

The definitions of the LAS communication protocol, messages, and expected behaviors are located in the LAS Interface Software section (Page 73 *LAS Interface Software*).

The serial communication signals are compliant with EAI/TIA232E CCITT V.28.



-
- 1 LAS DB9/Ethernet Interface panel location
 - 2 LIS DB9/Ethernet Interface panel location
-

The following table describes the D-sub 9 Connector Pinout:

Serial connection pinouts

Pin	Signal	Signal name	DTE (PC)
2	RXD	Receive Data	In
3	TXD	Transmit Data	Out
5	GND	Signal Ground	—

About LIS Connection Pinouts

The SHC system does not supply a serial or Ethernet cable in its kit. The operator must use the LIS connection cable. The Atellica Solution has a male DB9 and Ethernet interface on the back of the module. The location of the connector is displayed on the back of the system in the previous figure.

About Other Connections

The LAS communication cable is the only connection between an Atellica Solution and an LAS. An Atellica Solution and an LAS do not have the following connections:

- AC power
- DC power
- Safety ground
- Structural
- Mechanical
- Cosmetic mechanical

Physical Characteristics of the SHC

The following sections contain the physical and mechanical interface properties of the SHC module.

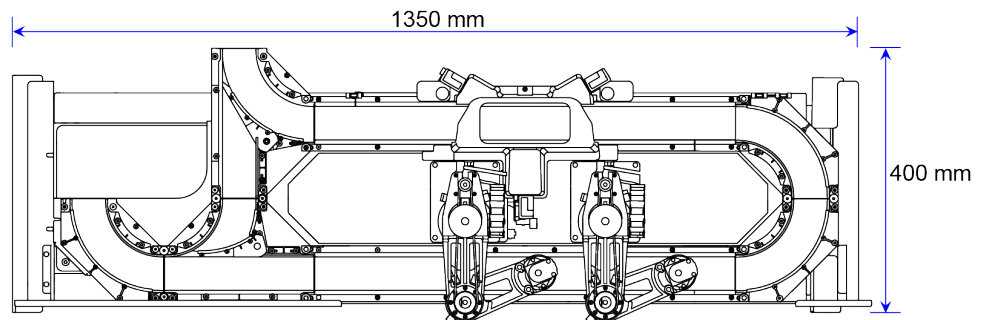
SHC Dimensions

The following table lists the physical dimensions of the SHC.

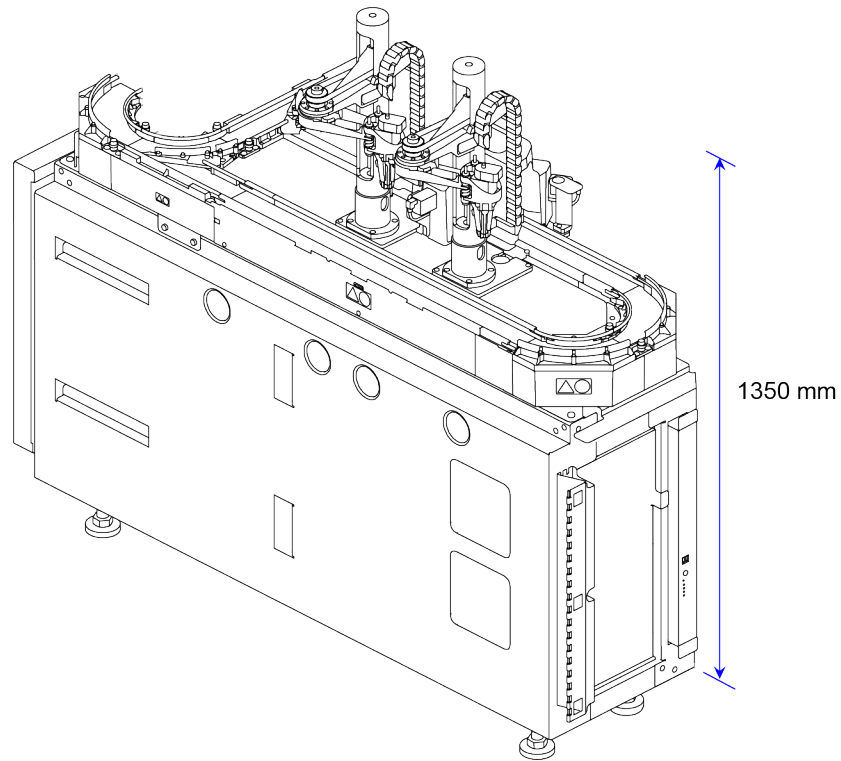
Item	Parameter	Value
Outside Measurements of Covers	Length (to lower doors)	1350 mm
	Length (to covers)	1430 mm
	Width	400 mm (not including LAS cover overhang)
	Height	1350 mm
Cover of Transfer Arm on the left side of the SHC (to take samples from the LAS buffer)	Length	145 mm (additional to the overall width of the frame)

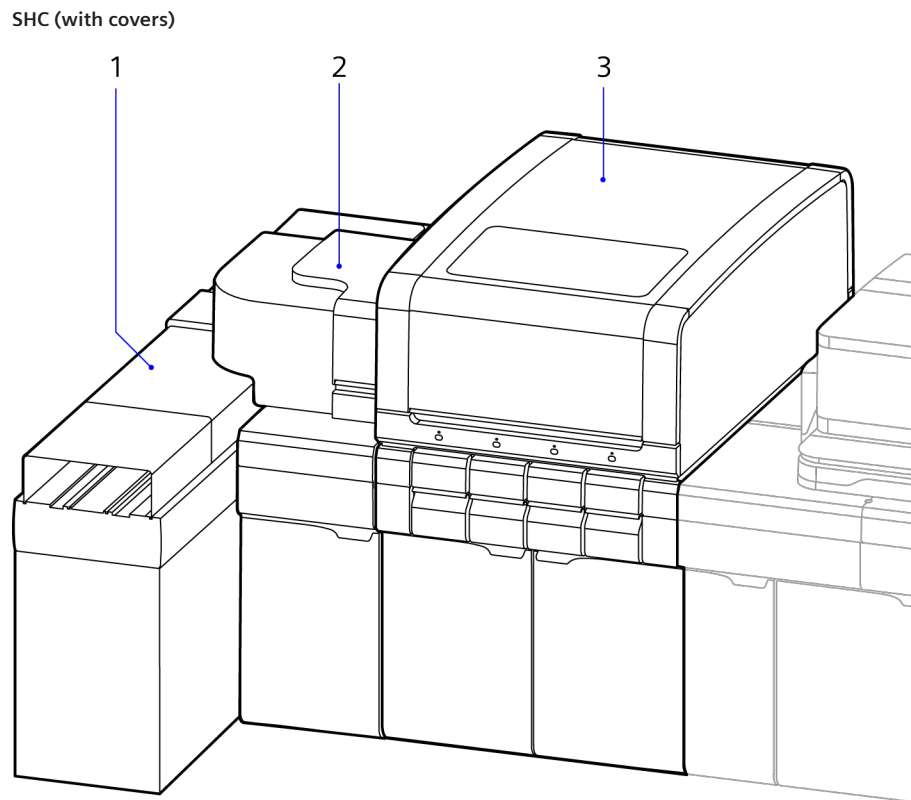
The following figures contain the applicable dimensions from the previous table respective to the intended orientation.

SHC interface top view



SHC (covers removed)



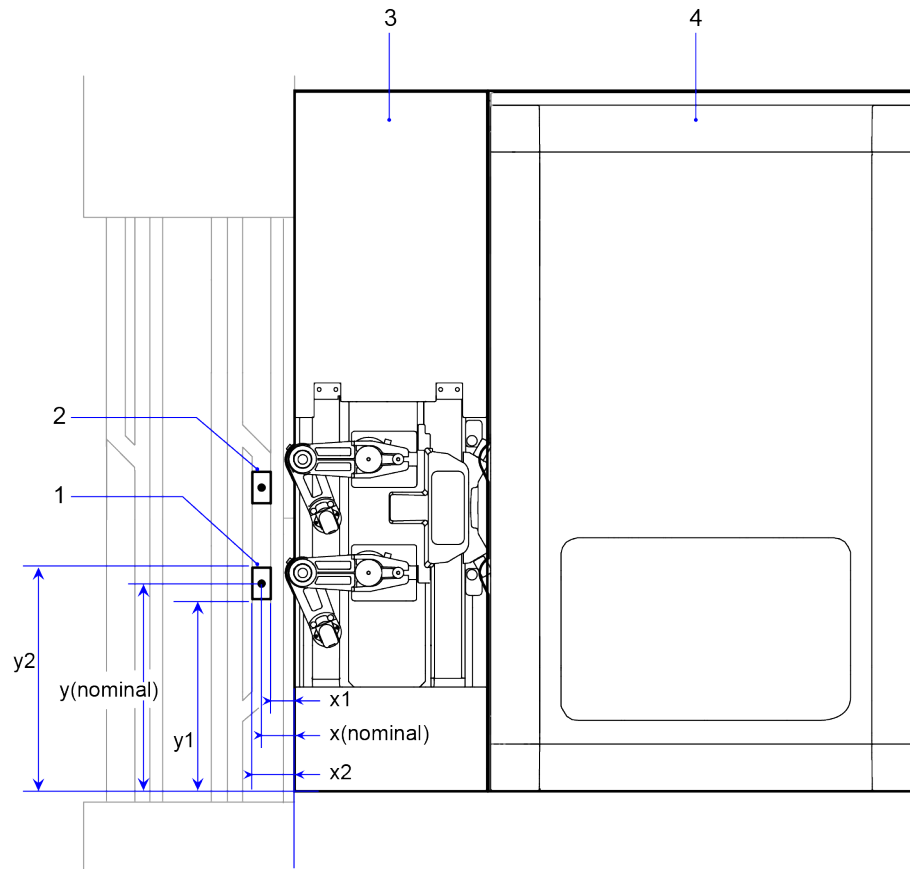


-
- | | |
|---|-------------------|
| 1 | LAS |
| 2 | SHC |
| 3 | Atellica Solution |
-

About Nominal Positioning

The following figure displays the nominal installation position relative to the LAS buffer. The required spacing between the LAS and SHC frame to remove the automation covers is a minimum of 10 mm and a maximum of 30 mm.

Nominal Positioning



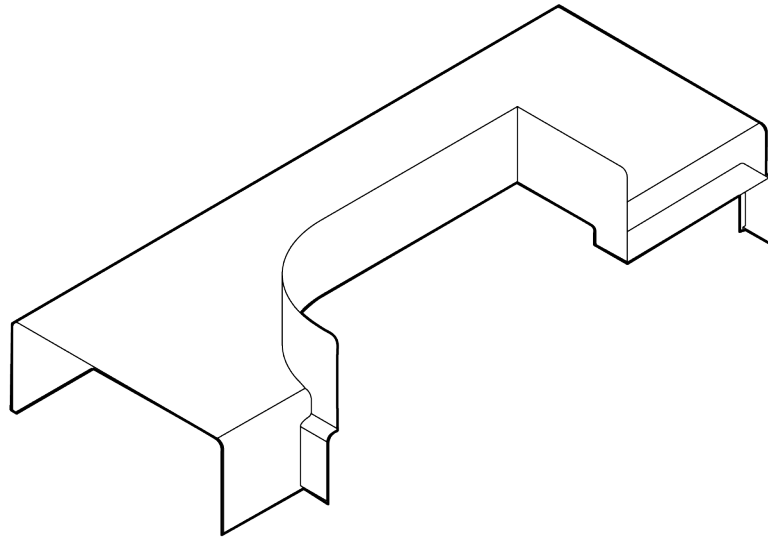
-
- 1 Pick Position
 - 2 Place Position
 - 3 SHC
 - 4 SH
-

Arm-1	y-nominal = 331 mm	y1 = 323 mm	y2 = 339 mm
Arm-2	y-nominal = 564 mm	y1 = 556 mm	y2 = 572 mm
Both Arms (to frame)	x-nominal = 96 mm	x1 = 91 mm	x2 = 101 mm
Both Arms (to SHC cover)	x-nominal = 82 mm	x1 = 77 mm	x2 = 87 mm

NOTE: A minimum of 10 mm must exist between the SHC frame and installed automation component.

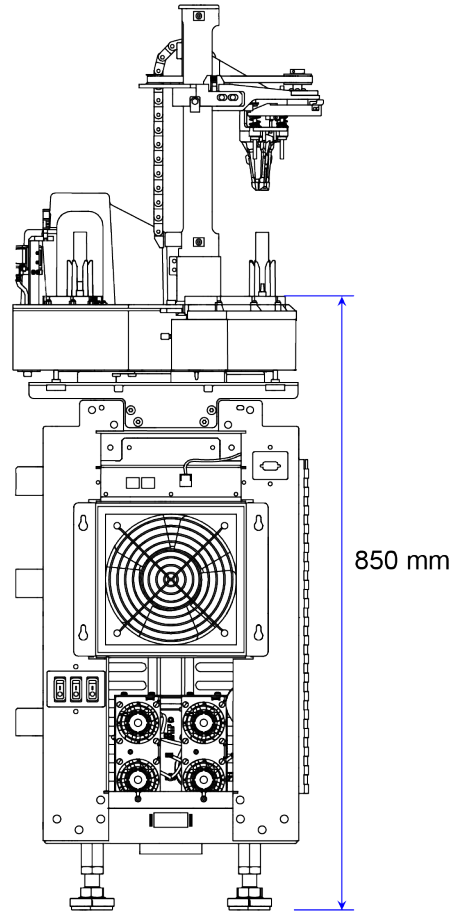
About Track Cover Opening

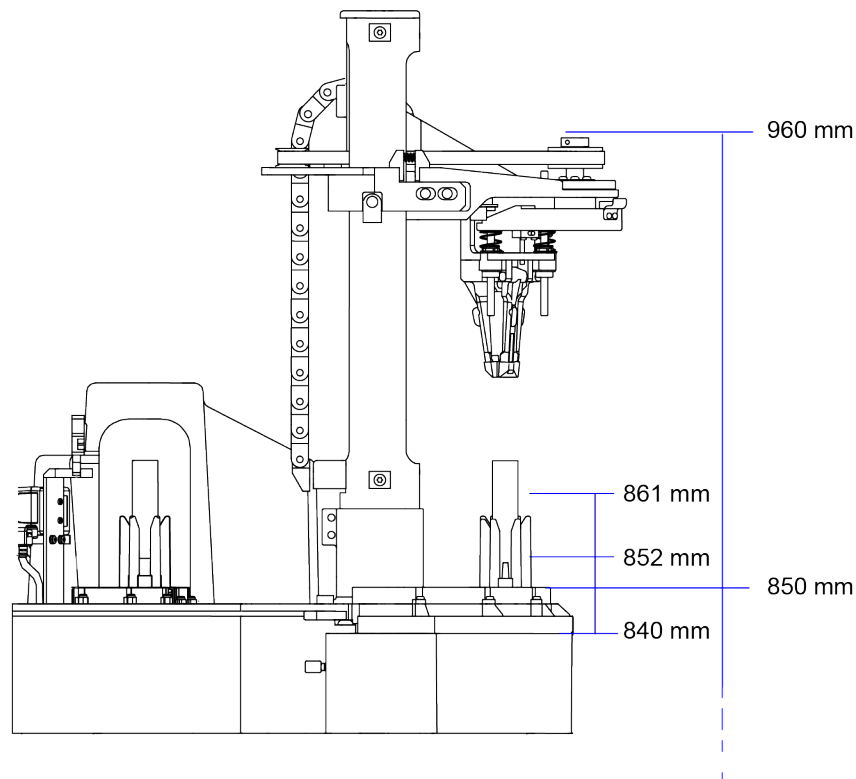
The LAS track cover opening provides access to an input and output position.



The opening has a vertical wall around the horizontal portion of the opening to prevent spills from entering uncapped sample containers on the LAS. The wall is sufficiently high to account for variation in height between the track and the Atellica Solution interface where the SHC cover overlaps.

About Interface Range of Motion





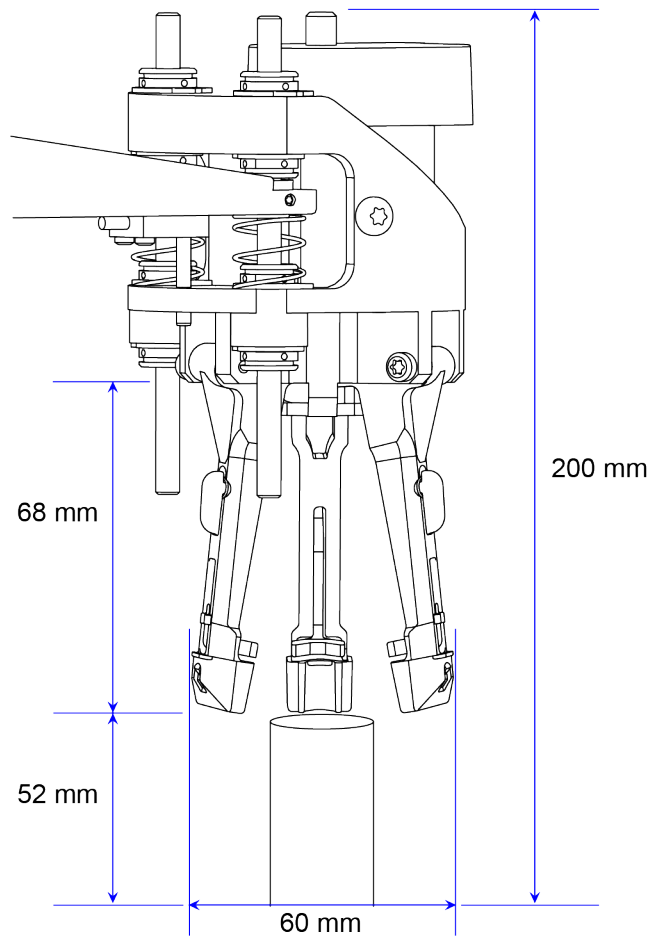
About Specific Requirements between SHC and Automation for Range in Motion

For timing and height compatibility, the arm does not attain the full vertical distance above a tube positioned in an adjacent carrier. As a result, the Atellica Solution sends the LOAD-UNLOAD COMMAND RESPONSE message until the SHC robot moves horizontally sufficiently to avoid tube-to-tube interference.

When the SHC robot is blocking tube movement in the buffer, the LAS communication protocol includes messaging to “freeze” the LAS buffer.

About Clearance around Tubes

The following figure describes the envelope and range of motion of the transfer gripper system. The gripper uses 3 electrically actuated fingers to grip and release sample containers.



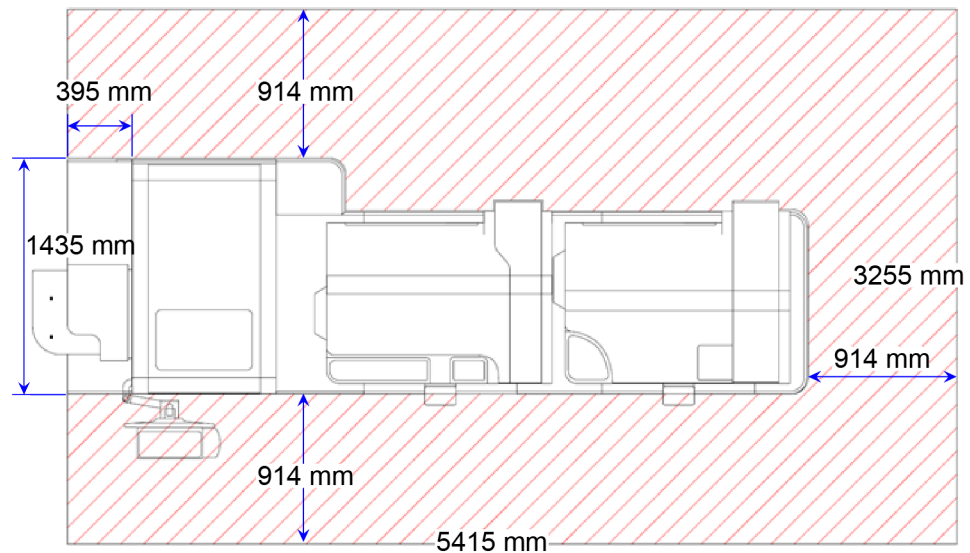
SHC components do not interfere with the largest capped tube in a pallet passing through the LAS buffer or adjacent to the Atellica Solution buffer pick-and-place position in the following scenarios:

- The inside LAS track path for any SHC operating condition. Inside track path is the bypass path in a typical diverter-type track connection.
- When the LAS has not sent a LOAD-UNLOAD COMMAND.

NOTE: SHC components include the SHC gripper, robot mechanism, and any tube in the SHC gripper.

About Service Envelopes

The following figure displays the approximate offsets for service clearance with the covers removed. The back of the Atellica Solution requires 1000 mm of clearance to enable service access points.



About Safety Cover Envelopes

The SHC safety cover has 3 covers:

- A primary cover enclosing the SHC transfer arms and transfer arm interface positions
- An Atellica Magline Transport cover protecting the back of the Atellica Magline Transport
- An operator access door

NOTE: All covers are removable by service personnel for access or repair of the Atellica Solution or LAS components. The cover interfaces with the opening in the LAS buffer covers.

About Seismic Anchoring

The lower cross members of the SHC frame has 4 anchor points. This allows permanent mounting in installation localities requiring seismic anchoring. The anchor points facilitate the attachment of third party seismic anchoring hardware.

NOTE: The Atellica Solution does not include dedicated anchoring hardware and is not designed for any specific third part mounts.

The contracted seismic engineer responsible for installation determines the required hardware based on building construction, flooring material, and local building codes. They are also responsible for ensuring the seismic anchoring hardware does not interfere with operation or service of the Atellica Solution or LAS.

About System Alignment

The SHC module is not a standalone piece of hardware and is intended to attach to the SH frame at installation. Alignment of the Atellica Solution to an LAS involves semi-automatic and manual steps. During an Atellica Solution installation to an LAS, the operator places the SH and SHC first. After placing and leveling the SH and SHC within the target location next to the LAS buffer, the operator any additional Atellica modules.

The installation procedure:

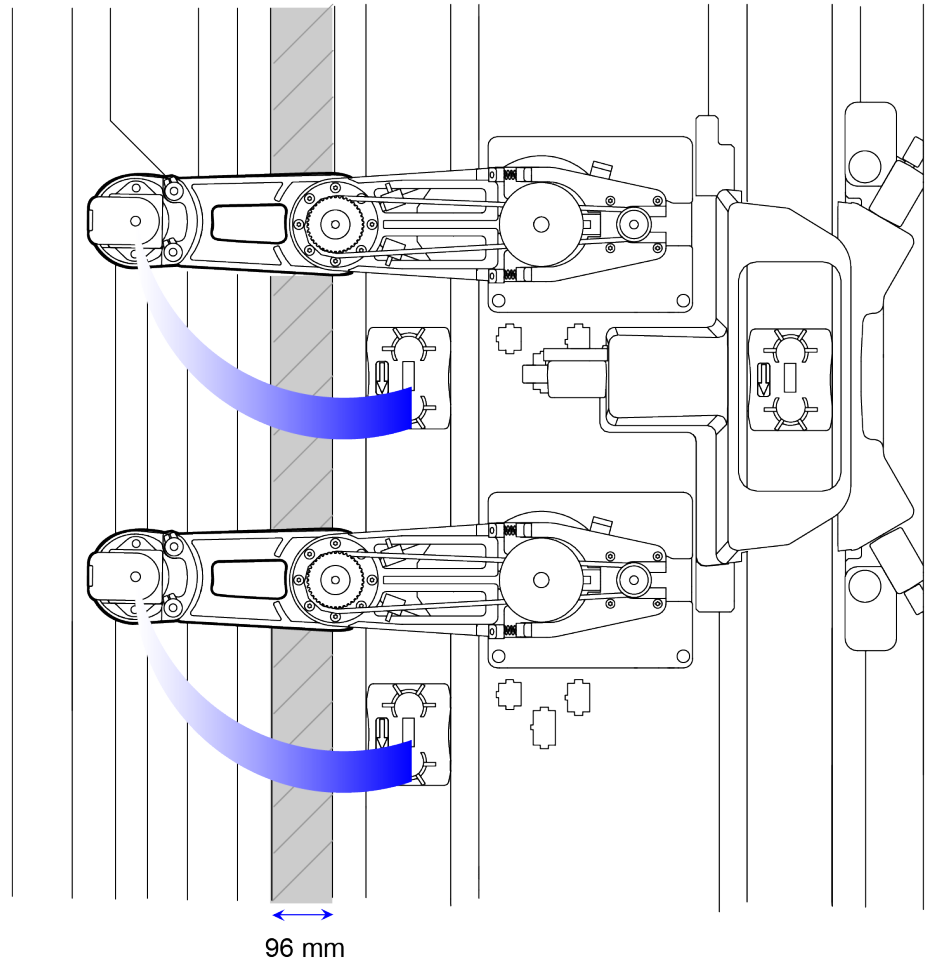
Installing the SHC Module

1. Attach the SH and SHC frames.
2. Position the SH with attached SHC offset from the Atellica Magline Transport by a minimum of 25 mm.
3. Align the linear position of the SHC to LAS buffer Interface Positions using the alignment marks on SHC panels.
4. Set the levelers to hold the respective SH or SHC position.
5. Perform the final Atellica Solution leveling prior to SHC robot alignment to the LAS buffer Interface Positions.
6. If appropriate, adjust the gate position on the LAS track or arm position.

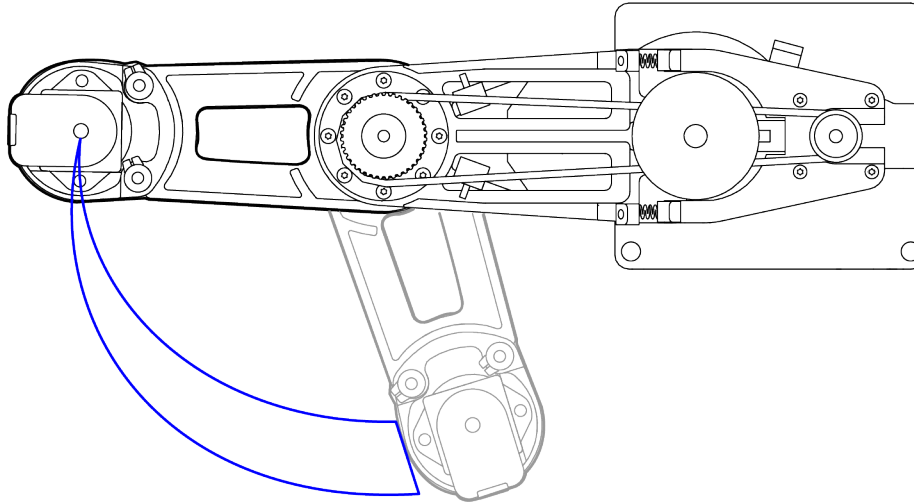
7. After operation commences, execute the teach sequence for arm positions.

NOTE: The teach sequence at the Automation Interface Positions is semi-automated.

SHC interface range of motion and adjustment (1/2)

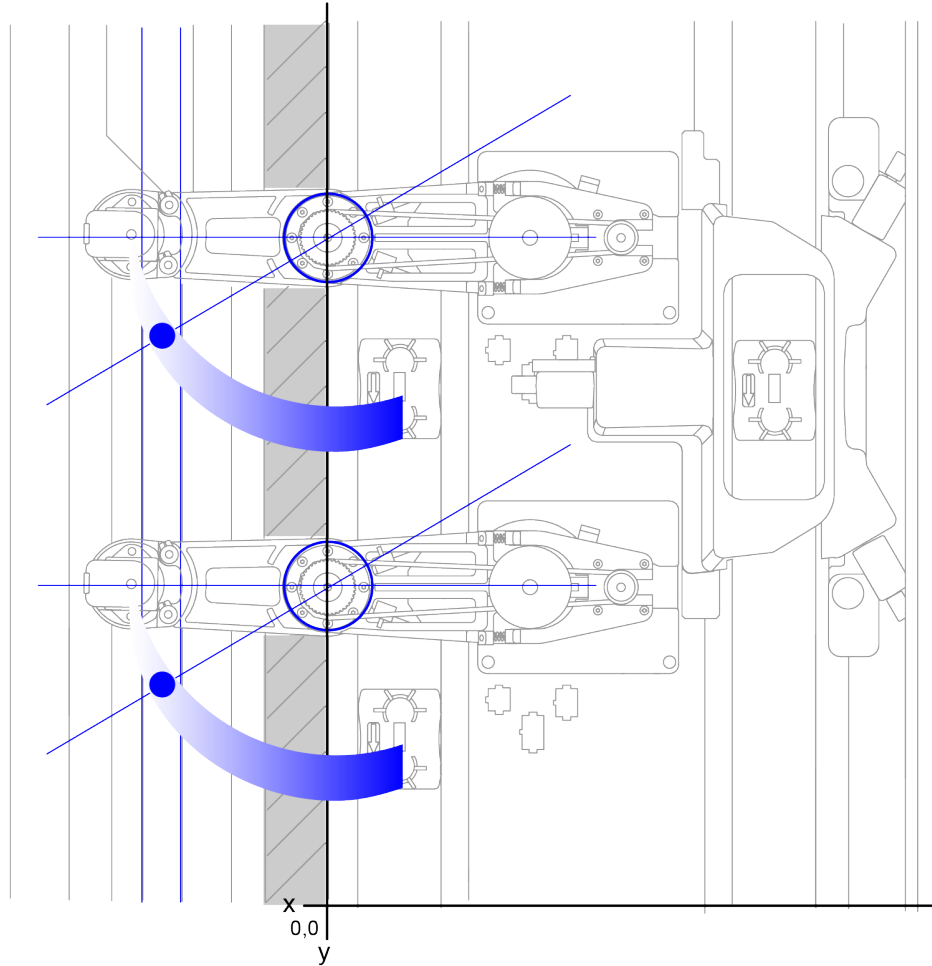


SHC interface range of motion and adjustment (2/2)



The following figure shows how the range of motion relates to the x and y positioning of the system. 0,0 indicates the front left corner of the SHC system (Page 58 *About Nominal Positioning*).

SHC schematic adjustment range



SHC Precondition

About Sample ID Read

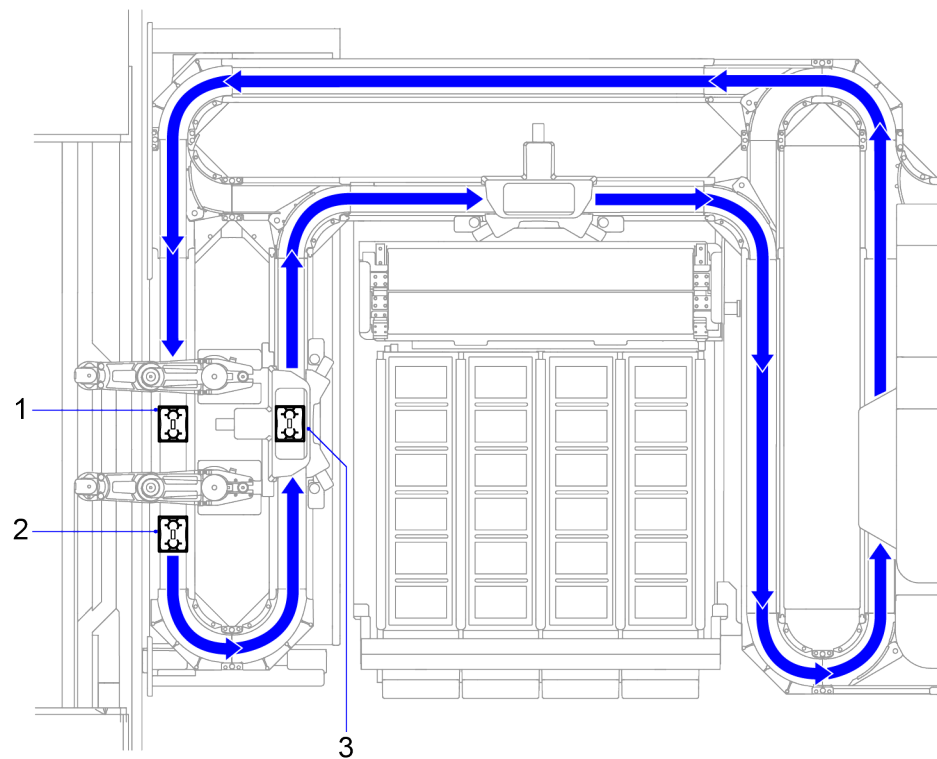
The LAS provides the sample container sample ID to the Atellica Solution in the LAS communication LOAD-UNLOAD COMMAND message. After acceptance from the LAS, the SHC transfers the tube onto the Atellica Solution. The TCS re-reads the barcode and re-confirms with the barcode reader on the SHC. Any mismatch between the LAS supplied ID and the Atellica Solution captured ID results in the sample returning to the LAS as incomplete with sample ID error.

The Atellica Solution uses the complete ASCII character set (including control characters) for barcodes.

A valid sample barcode is subject to the following restrictions:

- Minimum length: 4 characters
- Maximum length: 29 characters
- Do not use the following prefixes:

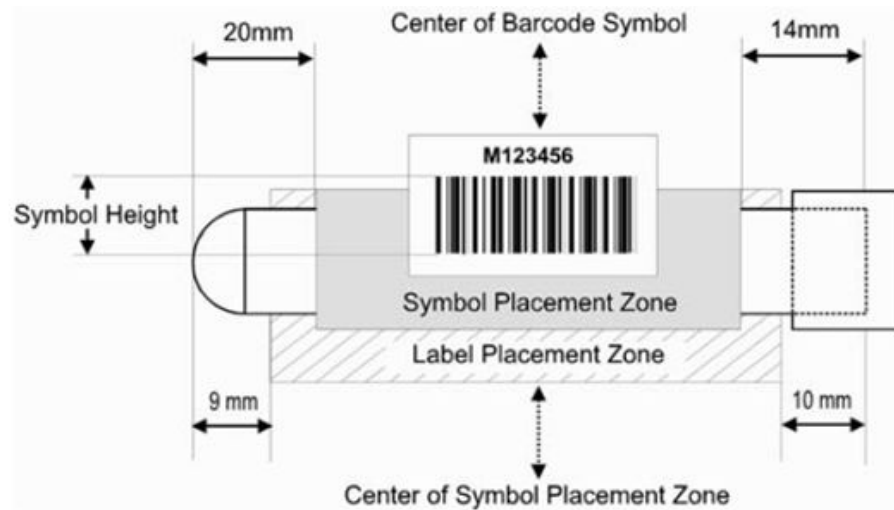
:>\$	Reserved for sample rack barcodes
:>%	Reserved for calibrator rack barcodes
:>&	Reserved for control samples



- | | |
|---|-----------------------|
| 1 | Transfer position |
| 2 | Transfer position |
| 3 | Barcode read position |

About Sample ID Placement

The symbol placement zone is dictated by the *Clinical and Laboratory Standards Institute Auto02-A2, Laboratory Automation: Bar Codes for Specimen Container Identification; Approved Standard—Second Edition*, Volume 25, Number 29. The following figure details the limits:



NOTE: The size of the placement zone varies depending on the length of the tube.

About Consumables

No consumables are directly related to SHC or LAS connectivity.

About Sample Processing LAS Requirements

After SHC robot alignment, the SHC robot grips sample containers at the LAS buffer pick position for variations in tube centerline position. The tube centerline lies within a cylindrical zone 2.1-mm diameter centered on the nominal pallet centerline over the height of the tube.

After SHC robot alignment, the LAS pallets allow sample container insertion for variations in tube centerline to pallet centerline position. The tube centerline is within a cylindrical zone 2.1-mm diameter centered on the nominal pallet centerline over the height of the tube.

When sample container to pallet alignment is within the 2.1-mm diameter cylindrical zone, no more than 10 Newtons of force is allowed when inserting a sample container into the carrier. The SH Pick-and-Place Station (PPS) produces a Z direction force larger than previously indicated.

NOTE: This force level applies to sample containers on the Atellica Solution list of approved sample containers.

The force to remove a sample container from a carrier is approximately 5 Newtons when sample container to robot gripper alignment is within the 2.1-mm diameter cylindrical zone defined above. The PPS robots are capable of producing a Z-direction force larger than indicated above.

NOTE: This force level applies to sample containers on the Atellica Solution list of approved sample containers.

5 LAS Interface Software

The Atellica Solution's LAS-Driver communicates with the LAS using a uRAP-based protocol, which includes messages to support sample transfer between LAS and an Atellica Solution. The LAS communication is for exchanging system event and alert information, assay inventory status, and sample exchange messages.

About LAS Communication Rules

The following are the general LAS communication rules (Page 82 *About Establishing LAS Connection*):

NOTE: The Atellica Solution uses a client–server model with the LAS acting as the client and the Atellica Solution acting as the server.

- Connection Made: When handshake messages exchange finishes, the communication connection occurs (Page 82 *About Establishing LAS Connection*).

- Keep Alive message transmission starts upon making connection.
- Except for locked carrier knowledge, the Atellica Solution resets all other information when communication is lost.

NOTE: “Locked carrier” refers to an LAS pallet at an LAS buffer Interface Position where the LAS sends a LOAD-UNLOAD COMMAND message and the connected Atellica Solution does not respond with the appropriate message. The LAS does not release the pallet from the Interface Position (pallet is locked in position) because the Atellica Solution sample container robot can move at any time after the Atellica Solution sends LOAD-UNLOAD COMMAND message. Indexing (unlocking) the pallet before the Atellica Solution sends the appropriate message can cause loss of sample and other hazards.

- Until communication occurs, the LAS only sends handshake messages and the Atellica Solution responds. This ensures messages are exchanged in working state. Additionally, this allows either side to reset the connection when there is a connection loss by failing ACK/NACK responses.
- After connection occurs, the connection reset rules apply (Page 107 *About LAS Connection Reset*).
- The connection initialization step requires a series of message exchanges that the LAS initiates. After connection is initialized, sample routing and sample transfer can begin, and the Atellica Solution begins transmitting status updates unsolicited.

Initialization message	Notes
Clear Queue	Clear queue does not affect previously locked carrier.
Onboard Sample Info Request and Response	—
Instrument Health Request and Response	If the operator receives the Instrument Health Response with a lock the Atellica Solution owns, then unsolicited Load_Unload Command Response is required before connection initialization completes.

Initialization message	Notes
Test Inventory Request and Response	Solicited response contains full test inventory information.
Consumable Inventory Request and Response	Solicited response contains full consumable inventory information.
Transfer Status Request and Response	—

- While it is permitted for LAS to query the Atellica Solution for information during initialization, the LAS does not poll the Atellica Solution because the Atellica Solution generates unsolicited response message as Atellica Solution status updates occur and solicited response messages may slow sample transfer throughput.
 - The Atellica Solution sends unsolicited responses when any changes occur.

NOTE: When LAS communication is initiated, the Atellica Solution does not accept any further initialization solicited requests. These solicited requests are NACKed with a Return Code of Message Type Not Supported (Refer to 0).

- In exception circumstances, if the LAS loses the test or consumable inventory data, the LAS resets connection to reestablish baseline.
- If the connection becomes ambiguous, disconnect and re-establish connection. For example, if a response to a command contains a mismatched barcode, the operator must reset the connection and re-baseline the connection.
- Only 1 interaction message can be active at a given time. Before the Atellica Solution sends the next command for Interface Position, the existing interaction must complete by response message and acknowledgment of response message. Only 1 of the following messages can be active per interface position:
 - Load_Unload Command request
 - Skip Command request
- The operator can send a message on Acknowledgment timeout, but cannot retry it if the response times out. If a response message times out, then the operator must reset connect.

- Sample ID support:
 - Maximum Length: 20 characters. LAS protocol messages supports 24 characters sent by LAS but the Atellica Solution does not accept samples with greater than 20 characters.
 - Supports Alphanumerics
 - Supports standard 7-bit ASCII symbols (for example, "!@#\$%^&*()"). Sample ID should not start with & or % because these characters have special usage on an Atellica Solution.
 - Does NOT support non-printable characters or extended ASCII symbols (8-bit)
 - Case sensitive
- Unsolicited Messages:
 - Messages that are sent unsolicited:

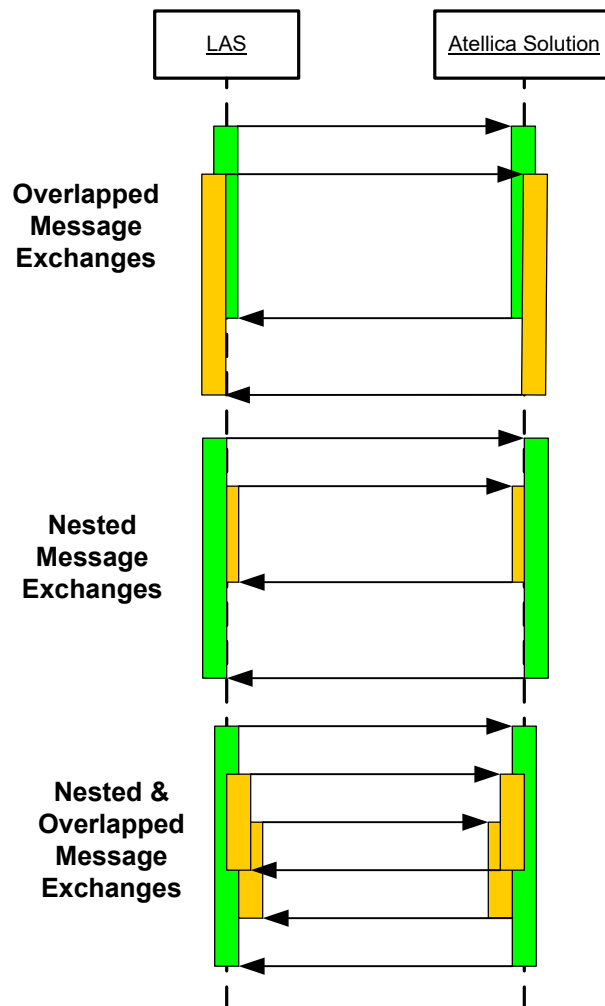
Unsolicited messages	Notes
Keep Alive message	[either side]
Transfer Status message	[Atellica Solution to LAS]
Onboard Sample Info	[Atellica Solution to LAS]
Instrument Health	[Atellica Solution to LAS]
Test Inventory	[Atellica Solution to LAS]
Consumable Inventory	[Atellica Solution to LAS]
Load Unload Command Request messages	Load Unload Command Request messages have higher priority than unsolicited messages as they directly affect sample throughput.
Test Inventory	—
Consumable Inventory	—
Onboard Sample Info	—
Load Unload Command Response messages	Load Unload Command Response message have lower priority than unsolicited Instrument health messages to ensure that LAS is aware of the Atellica Solution health prior to lock release.

Unsolicited messages	Notes
Keep Alive messages	(Page 110 <i>About Keep Alive Messages</i>).
Transfer Status Message	(Page 120 <i>About Transfer Status Messages</i>).
Onboard Sample Info	When the operator removes any LAS sample from the Atellica Solution without unloading it to the track, an unsolicited message displays. This message displays when the Atellica Solution recognizes a positive sample loss event.
In case the operator removes a sample from the Atellica Solution and they cannot report the removal to the LAS, the operator is responsible to inform the LAS about the sample removal. The operator cannot report the information in the following conditions: • If the connection between the Atellica Solution and the LAS is lost. • If the sample containing the Atellica module is offline or not communicating.	—
Instrument Health Update on Status change	—
Unsolicited messages generated for following status changes:	—
LAS Interface Status	—
Instrument Process Status	—
Remote Control Status	—
LIS Connection Status	—
The frequency of these unsolicited response messages is gated to limit traffic.	—

Unsolicited messages	Notes
A configurable delay between unsolicited Instrument Health Response messages is provided: (Default: 5 seconds; Range: 5–60 seconds).	—
Test Inventory Update on Status change.	—
During LAS connection initialization, the Atellica Solution transmits full test inventory in a single message.	—
The unsolicited message only contains the Test Inventory that has changed since the last response message transmission.	—
The Atellica Solution generates unsolicited messages on any reagent status change or reagent level change of the configurable step size: • Default: 22 count • Range: 5–100	A large number of reagents changing to Red or to Green from Red is an exception: • Change in consumable availability. • Atellica module becoming active.
The Atellica Solution gates the frequency of this unsolicited response message to limit traffic.	A configurable delay between unsolicited Test Inventory response messages: • Default: 5 seconds • Range: 5–60 seconds
Consumable Inventory Update on Status change.	—
During LAS connection initialization, the Atellica Solution transmits full consumable inventory in a single message.	—
The unsolicited message only contains the Consumable Inventory that has changed since last response message transmission.	—
The Atellica Solution generates unsolicited messages for any consumable status change.	A large number of consumables changing to Red or to Green from Red is an exception: • Change in consumable availability. • Atellica Module becoming active.

NOTE: If the Atellica Solution receives a request or response message with an invalid value, the message is Nacked.

- The message exchange can be nested, overlapped, or both.
 - The operator cannot issue multiple commands on a sample interface gate pallet.



About uRAP Protocol Primitives

All uRAP messages are in a uRAP protocol Header & Footer. Transportation protocol is not part of the uRAP protocol Header & Footer.

About Field Length

A Field Length byte (1 byte) prefixes a dynamic length byte field (ASCII string field), such as sample ID. When a given field length value is 0x00, the dynamic length byte field is not present. The maximum value of the field length is 0xFF.

Headers

Header format	Size (bytes)
Start Signature	1
Message Length	2
Sequence ID	2
Return Sequence ID	2
Message Type	2
Time Stamp	8
Instrument ID	1

About Start Signatures

The Header starts with an <stx> byte. This value, which is the standard starting byte value for most serial communication protocols, is present regardless of the employed transport mechanism.

NOTE: When the Atellica Solution employs serial communication as the transport mechanism, the <stx> byte appears twice in each message. The application layer inserts 1 instance of the byte, and the transport layer inserts the other instance.

About Message Length

The message length specifies the number of bytes in the entire message, including the header and footer. The message length does not include any header or footer bytes added by the transport layer.

About Sequence ID

The sequence ID is a unique identifier for ordering and keeping track of the messages. Each sender, such as the Atellica Solution and the LAS, performs their own sequence ID counter.

The sequence ID starts at 1, and the sequence ID of 0x0000 is reserved for use in Return Sequence ID. When sequence ID reaches maximum value (0xFFFF), it is rolled over to 0x0001. The Atellica Solution resets the sequence ID value upon successful handshake.

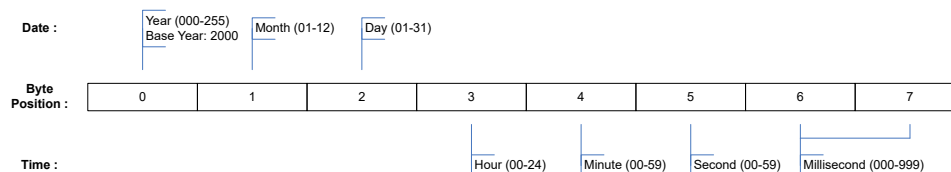
About Return Sequence ID

The return sequence ID indicates if the message is a response message or an unsolicited message. If the return sequence ID is empty (0x0000), then the message is an unsolicited message. If the return Sequence ID is occupied with a non-zero sequence ID, then the message is a response message. The operator must set the return sequence ID to the sequence ID of the received message when it is a response or ACK/NACK message.

About Message Types

The message type is a 2-byte binary code to indicate which type of command or response message it is.

About Time Stamps



The timestamp shows message origination date and time using local date and time. The bytes are arranged in format of largest gradient of time to smallest for sorting purposes. The custom format arrangement maintains neutrality to any system date and time format of any operating system and programming language. The baseline time is January 1st, 2000. If the operator cannot set the time, they must set the value to the following: Year: 0, Month: 1, Day: 1, Hour: 0, Minute: 0, Second: 0, Millisecond: 0.

About Instrument ID

The instrument ID is an enumerated number for the device on a communication network (Page 84 *About Handshake Messages*).

Footers

Parameter	Size (bytes)
Checksum	2
End Signature	1

About Checksums

The checksum field is always present, even if the transport layer protocol has its own checksum. This requirement follows from the goal of making the URAP application layer independent of the transport layer.

The Atellica Solution calculates the checksum value from uRAP header and message body (not including the footer section). The algorithm is a binary sum, formatted as the ASCII representation of its hexadecimal value (modulo 256) in 2 digits with leading zeroes, a range of 00–FF.

About End Signatures

The Footer ends with an <etx> byte. This value, which is the standard ending byte value for most serial communication protocols, is present regardless of the employed transport mechanism.

NOTE: When the Atellica Solution employs serial communication s the transport mechanism, the <etx> byte appears twice in each message. The application layer inserts 1 instance of the byte, and the transport layer inserts the other instance.

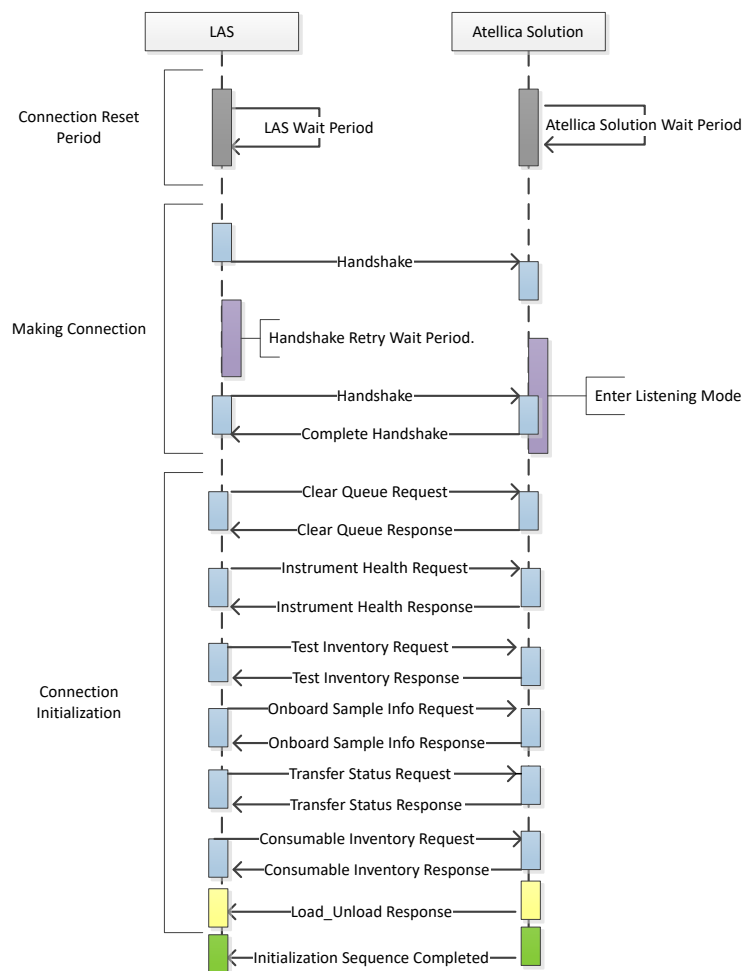
About Establishing LAS Connection

To establish a connection, the client-server model uses the Atellica Solution as the server and the LAS as the client. When the Atellica Solution powers on and initializes its software, it enters into listening mode and monitors the LAS communication port for handshake message.

The LAS as client is responsible for initiating the handshake message exchange. The Atellica Solution responds with an acknowledgment message to the LAS handshake request and replies with a handshake response message. The LAS considers the connection complete when it receives the handshake response message from the Atellica Solution. The Atellica Solution considers the connection complete when it receives acknowledgment for its handshake message from the LAS.

NOTE: The operator can send Initialization Request messages in any order. If the Instrument Health Response reports 1 or more, the Atellica Solution locks the IP and the LAS must wait for respective IP's unsolicited response or responses to complete Connection Initialization. The Atellica Solution waits for acknowledgment for those responses to complete Connection Initialization.

The LAS waits a maximum of 600 seconds for a Load_Unload response message.



Serial Port Settings

LAS communication uses the serial port. The following settings define available configuration values and respective default configuration values:

Connection parameter	Valid values	Default value
Com port	COM1 COM2 COM3 COM4	Hardware Specific
Baud rate	9600, 19200, 38400, 57600, 115200	57600

Connection parameter	Valid values	Default value
Flow control	None	None [uRAP module does not use Flow Control for a serial connection]
Parity	None, Odd, Even, Mark, Space	None
Data bits	7, 8	8
Stop bits	1, 2	1

About Handshake Messages

The handshake message does not follow a typical request or response message pair format. Single message format exists for use by the LAS and Atellica Solution.

Header	Protocol Version (2 Bytes)	Instrument Type (2 Bytes)	Capability Version (2 Bytes)	Software Version (2 Bytes)	Instrument ID (1 Byte)	F L	Instrument Serial # (n Bytes)	Footer
--------	-------------------------------	------------------------------	---------------------------------	-------------------------------	---------------------------	--------	----------------------------------	--------

Handshake message format and values	Notes
Message Direction	Bi-Directional
Message Type	0x0001
Protocol Version	The uRAP protocol version is divided: • High Byte: Major Version • Low Byte: Minor Version
Instrument Type	The enumeration of the instrument types.
Implemented Capability	The enumeration of the capabilities of the instrument or automation system using this protocol version: • High Byte: Capability (for example, Core = 0x01, Enhanced = 0x02, Advanced = 0x03, and so on)

Handshake message format and values	Notes
Software Version	The software version is filled with 0xFFFF.
Instrument ID	The Atellica Solution and the LAS utilize FFh to indicate Instrument ID.
Instrument serial #	The instrument serial # is for verification of the instrument.

The following specific implementation adjustments apply to simplify handshaking:

- The Atellica Solution uses Software Version number 1.0.
- The Atellica Solution uses 0xFFh and Instrument ID is disabled.
- Instrument Serial # is fixed, and automation does not transmit the expected serial number.

NOTE: Handshake Messages is NACKed if the operator uses invalid parameters.

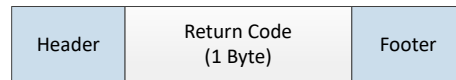
Handshake	Used by	Fixed values
Protocol version	Both	0x0330h
Instrument type	Both	Instrument = 0x0001h LAS = 0x0000h
Implemented capability version	Both	0x0104h
Software version	LAS	0xFFFFh
Software version	Instrument	0x0100h
Instrument ID	Both	0xFFh
Instrument serial #	Both	FL set to 8. "ATELLICA"

Handshake message configuration item	Default values
Handshake Retry Period	30 seconds (range 10–300)
Handshake Response Timeout	20 second (range 2–60)

Handshake message configuration item	Default values
Wait Period Prior to Initiating Handshake (LAS)	30 seconds (range 10–600)
Wait Period Prior to Entering Listening Mode	15 seconds (range 10–600)
Wait Period to complete initialization sequence after Handshake (LAS)	30 seconds (range 10–600)

Acknowledgment Message

The Atellica Solution sends a Message Acknowledgment (ACK/NACK) is sent upon each message it receives. The ACK/NACK message uses 0x0000 as the message type:



Where Return Code may be 1 of the following values:

Acceptance state	ACK/NACK	Value (Hex)	Usage
Message Understood & Accepted	ACK	0x00	- Properly formed Atellica Interface Specification message. - Request messages received during connection initialization. - Non request message received after connection initialization.

Acceptance state	ACK/NACK	Value (Hex)	Usage
Message Not Understood	NACK	0x01	• Message contains unexpected enumeration. • Illegal composition of message fields such as mismatched field data. – Example 1: Number of test inventory do not match actual inventory data. – Example 2: If carrier occupancy do not match with barcode entry. – Example 3: Message type is known but internal field does not match with known message structure.
Message Type Not Supported	NACK	0x03	• Any message the Atellica Solution receives prior to handshake. • Request message after connection initialization. • Any message that the Atellica Interface Specification does not support.

About Initializing the LAS Connection

After the systems exchange and ACK handshake messages, the LAS sends the following 6 requests to initialize the connection:

- Clear Queue Command Message
 - This message explicitly clears any previous queue in the Atellica Solution memory. However, any previously locked carriers are not cleared by this message.
- Onboard Sample Info Request Message

- Instrument Health Request Message
 - In the response message, if the LAS Interface Status is Critical (0x04), the LAS does not move any pallets in the LAS buffer as the Atellica Magline Transport mechanism or mechanisms may be obstructing the pallet or sample container travel path (Page 97 *About Automation Interface Status: Critical*).
 - If the LAS owns the Instrument Health Response with a lock the Atellica Solution owns, then an unsolicited Load_Unload Command Response is required before connection initialization can complete.
- Test Inventory Request Message
 - Any solicited Reagent Inventory response message include full reagent inventory information.
 - Any unsolicited reagent inventory response messages only include changes from previous response message (Page 73 *About LAS Communication Rules*).
- Consumable Inventory Request Message
 - A solicited Consumable Inventory response message includes full consumable inventory information.
 - An unsolicited Consumable Inventory response message only includes changes from the most recent response message (Page 73 *About LAS Communication Rules*).
- Transfer Status Request Message
 - A Transfer Status Request or Response exists for each interface position (IP0 and IP1).

After handshake messages are exchanged and ACK'd, the LAS waits for the following message to make sure the initialization is complete:
- Initialization Sequence Complete Message
 - This unsolicited Initialization Sequence Complete message indicates the completion of all message exchanges.

The Atellica Solution does not transmit unsolicited messages until the LAS connection initializes. However, even after initializing the connection, carrier routing does not occur if following conditions are not met:

Loaded carrier routing

Element	Condition
Test inventory	At least 1 assay with usable test count

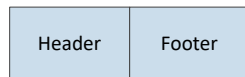
Element	Condition
Instrument health – automation interface status	Green (Can transfer sample tubes)
Instrument health – instrument process status	Green (Can process sample)
Instrument health – LIS Connection status	Connected
Instrument health – remote control status	IP0: Exchange Mode
Instrument health – remote control status (SHC)	IP0: Loading Only Mode IP1: Unloading Only Mode
Empty carrier routing	
Element	Condition
Instrument health – automation interface status	Green (Can transfer sample tubes)
Instrument health – remote control status	IP0: Exchange Mode or IP0: Unloading Only Mode
Instrument health – remote control status (SHC)	IP0: Loading Only Mode IP1: Unloading Only Mode or IP0: Offline IP1: Unloading Only Mode

The LAS must wait for all conditions to be met. The Atellica Solution sends unsolicited messages when its status changes.

Instrument Health Message

The Instrument Health Message informs the LAS about the condition (readiness) of the Atellica Solution and allows the LAS to make routing decisions. The Atellica Solution sends this data to the LAS in the Instrument Health Response message after receiving an Instrument Health Request message from the LAS. The Atellica Solution sends an unsolicited Instrument Health Response if any of the data values change.

NOTE: Atellica module identifiers (module serial numbers) do not include the following characters: \ ^ | : ; .

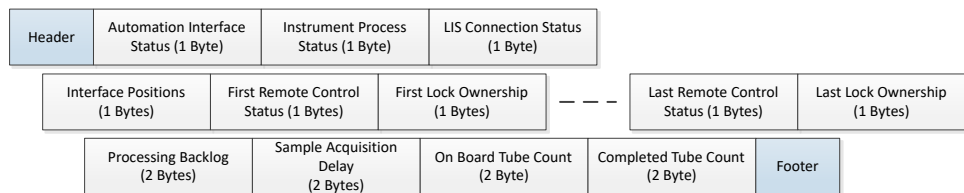


Instrument Health Request message

Notes

Message direction LAS to Atellica Solution

Message type 0x0201



Instrument Health Response message

Notes

Message Direction Atellica Solution to LAS.

Usage The Atellica Solution replies with LAS interface, Atellica Solution, Remote Control Status, and other parameters.

Message Type 0x0202.

Instrument Health Response message	Notes
Automation Interface Status (Page 97 <i>About Automation Interface Status: Critical</i>).	A software component monitors and controls the Transfer Arm and Transfer Vessel state. The Atellica Solution reports the status of the Transfer Arm and the Transfer Vessel to the LAS to indicate if the interface to the LAS is usable. Sample processing is only possible if the Transfer Arm and the Transfer Vessel are in working condition. Interference by the Transfer Arm may block pallet flow through the LAS buffer (Page 97 <i>About Automation Interface Status: Critical</i>). The Atellica Solution reports the Automation Interface Status in the following conditions: • 0x01 – Green: Transfer mechanism is fully functional and is not interfering with Track. • 0x03 – Red: Transfer mechanism is not functional and is not interfering with Track. • 0x04 – Critical: Transfer mechanism is not functional and is interfering with Track. Operator or Service intervention required.
Instrument Process Status	Message to report the process status of the Atellica Solution. For this status, the following is considered: NOTE: The LAS cannot change the Atellica Solution status using uRAP commands. Events within the Atellica Solution automatically trigger the status changes, such as start up, shutdown, consumption of consumables, or error conditions. 0x01 – Green: The Atellica Solution can process samples and has sufficient reagents and consumables. 0x02 – Yellow: One or more analyzer modules in the system configuration is not processing samples. 0x03 – Red: The Atellica Solution cannot process because of an error state or vital consumable depleted.

Instrument Health Response message	Notes
Affected Modules Count	When instrument processing status is Yellow or Red, the operator must provide the module and the reason for its unavailability. This field reports the number of modules currently not available. The maximum number of supported analyzers in the deployment is 7 as per the Atellica architecture.
Module ID	Affected Analytical Module's Serial Number. The maximum number of characters in the serial number is 15.
Module Not Available Reason	The reason that the Atellica Solution is not available. The system reports the following prioritized causes to the LAS: 0x01 – Operator initiated activity 0x02 – Consumable not available • A consumable or supply is not available. 0x03 – Empty waste bin • The operator must change the waste bin due to a continuous test. 0x04 – Initiate weekly maintenance • The operator must initiate maintenance activities due to an internal system behavior. 0x05 – Maintenance in progress 0x06 – Hardware error • The Atellica Solution has encountered a hardware error, such as a probe error. 0x07 – Magline error • Overall status is Yellow or Red due to the Atellica Magline status. 0xFF – Software Error • Default message if the error does not map to any of the defined causes.

Instrument Health Response message	Notes
LIS Connection Status	Message to report the current LIS connection status. The LIS connection status indicates the ability the Atellica Solution to download orders from the LIS. When the LIS is disconnected, then automation does not divert sample tubes to the Atellica Solution. • 0x01 – Connected • 0x02 – Disconnected
Interface Positions	The count of interface positions. SHC uses value 2. For each interface position, there is Remote Control Status and Lock Ownership.

Instrument Health Response message	Notes
Remote Control Status (Active Configuration of Interface Positions)	Remote control status indicates if each interface position accepts samples from LAS. NOTE: Interface Position Array starts directly behind the divert gate. The Atellica Solution returns the Remote Control Status in a single byte for each interface position in the LAS buffer. The preceding Field Length indicates the number of Interface positions. Interface Position 0 (IP0) is closest to sample entry position of LAS buffer. • 0x01 – Offline or Local Mode: Operator or system logic has Automation Interface offline and does not accept samples from the LAS. • 0x03 – Online Exchange Mode • 0x04 – Online Loading Only Mode • 0x05 – Online Unloading Only Mode NOTE: The Atellica Solution does not use all enumeration values per instance of system connection (Page 96 <i>About Remote Control Status</i>). The operator determines the Remote Control Status configuration based on the state of the SHC. The SHC can be 1 of the following states: • Online • Offline • Processing • Diagnostics

Instrument Health Response message	Notes
Lock Ownership	This field is for initializing connection. Upon each connection, LAS refreshes Lock Ownership from the Atellica Solution. For example, if the connection drops after the LAS has transmits a Load_Unload Command Request without receiving an ACK response, then it is ambiguous if the lock is currently owned by the Atellica Solution. • 0x01 – Locked by Instrument • 0x02 – Not Locked by Instrument NOTE: During LAS communication initialization, if the Lock Ownership status is 0x01 – Locked by Instrument for any Interface Position, then LAS communication is not initialized until the LAS receives an unsolicited Load_Unload Command Response message for all locked positions. After LAS communication initialization, the LAS does not request Lock Ownership status.
Processing Backlog	Not Applicable: The field exists in the message, but the values are unusable. The Atellica Solution populates this field with 0xFFFFh.
Sample Acquisition Delay	Not Applicable: The field exists in the message, but the values is not utilized. The Atellica Solution populates this field with 0xFFFFh.
On Board Tube Count	Number of tubes onboard from the LAS.
Completed Tube Count	Number of tube onboard ready to be return to the LAS.

Configuration items	Default values
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Instrument Health Response Timeout	20 second (range 2–60)
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About Remote Control Status

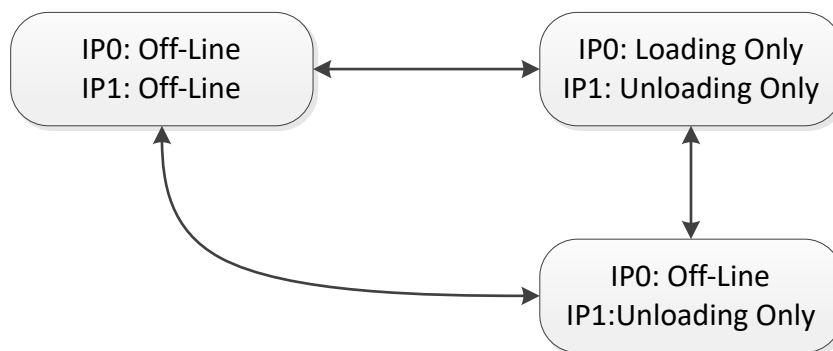
The Remote Control Status field of the Instrument Health Response message describes interface position's current active mode. In the Atellica Solution, the SHC is the 1 interfacing system.

NOTE: The Remote Control Status value usage may differ depending on the capability.

The SHC has 2 Interface Positions: IP0 and IP1. The operator typically uses IP0 to load pallets from the LAS buffer onto an empty Atellica Magline Transport on the system. The operator typically uses IP1 to unload sample carriers from the Atellica Solution back to the LAS.

Supported remote control by mode of implementation	Usable values
SHC (Interface Position 0)	0x01: Offline or Local 0x04: Online Loading Only Mode
SHC (Interface Position 1)	0x01: Offline or Local 0x05: Online Unloading Only Mode
DC supported status detail	Notes
IP0: Offline	When the Atellica Solution is configured to not process LAS samples and only process front loaded samples, IP0 is set to Offline.
IP0: Exchange	When the Atellica Solution accepts LAS samples normally, IP0 is set to Exchange.
IP0: Unloading Only	The Atellica Solution mode may change to return all LAS samples and not process any additional LAS samples. In this case, IP0 is set to Unloading Only.

SHC state transitions



SHC supported status detail	Description
IP0: Offline IP1: Offline	When the Atellica Solution is configured to process front-loaded samples in place of LAS samples, IP0 and IP1 are set to Offline.
IP0: Loading Only IP1: Unloading Only	When the Atellica Solution accepts LAS samples, IP0 is set to Loading Only and IP1 is set to Unloading Only. Initially, when the Atellica Solution is loading samples from the LAS and has no samples to unload, "Excess empty puck is cycled out of IP1 by Instrument" is displayed.
IP0: Offline IP1: Unloading Only	The Atellica Solution mode may change to return all LAS samples and not process any additional LAS samples. In this case, IP0 is set to Offline and IP1 is set to Unloading Only.

About Automation Interface Status: Critical

At times, the transport mechanism can fail because of a power outage or a motor jam. This obstructs the LAS pallet travel path. In these cases, it becomes necessary to halt the SHC modules and the LAS buffer and, alert the operator for error recovery to reduce the chance of contamination and loss of sample.

When the Atellica Solution recognizes this condition, it halts all its movement and sends an unsolicited Instrument Health Response. Automation interface status is set to Critical to report to the operator via its primary UI.

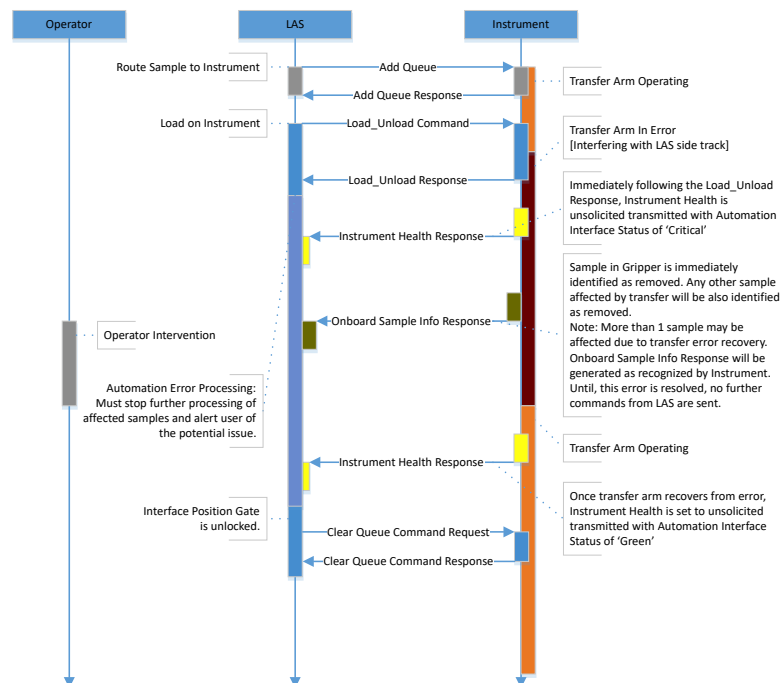
Except during initialization, the LAS does not send Clear Queue Command Request when the Automation interface status is Critical. During initialization, the sending of this message by the LAS is dependent on the automation interface status.

When the transport mechanism becomes usable or it does not obstruct the LAS pallet travel path, the Atellica Solution sends an unsolicited Instrument Health Response with the LAS interface status other than Critical.

The LAS may send the clear queue message for both interfaces and rebuild the logical queue by sending Add Queue commands- (in the order of the carriers that are physically in the queue) after it receives an Automation interface status other than Critical.

In establishing a connection, if the LAS receives a first Instrument Health Response message with:

- Automation Interface Status of Critical, the LAS does not move any pallets and waits for another Instrument Health response message with Automation interface status other than Critical.
- Lock Ownership status of Locked by Instrument, the LAS does not move any pallets and waits for a Load_Unload Command Response with Load Command Status not 0x02 and Unload Command Status not 0x02.



About Test Inventory Messages

Test Inventory messages inform the LAS about the assay portfolio and test availability (test counts) on the Atellica Solution. During Connection Initialization, the Atellica Solution sends the complete test inventory to the LAS via the Test Inventory Response message after receiving a Test Inventory Request message from automation. Automation can request complete test inventory only during connection initialization. The Atellica Solution periodically sends unsolicited updates that contain only those tests with status that have changed (Page 73 *About LAS Communication Rules*).

The response message contains only immediately usable test counts.

Header	Footer
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Test inventory request message	Notes
Message direction	LAS to Atellica Solution
Message type	0x0203

Header	Test Inventory Count (2 Bytes)	F L	First Assay Name (n Bytes)	First Usable Test Count (2 Bytes)	First Assay Status (1 Byte)
		F L	Last Assay Name (n Bytes)	Last Usable Test Count (2 Bytes)	Last Assay Status (1 Byte)
					Footer

Reagent inventory response message	Notes
Message direction	Atellica Solution to LAS
Usage	The Atellica Solution sends a message upon request or sends it unsolicited to indicate a change in reagent status. If responding to a Test Inventory Request, the Atellica Solution replies with reagent level for all active assays. Only reagent levels that have changed since the last response are included in unsolicited responses.
Message type	0x204.

Reagent inventory response message	Notes
Test inventory count	Number of test inventory counts in the message. This is the total number of tests on the Atellica Solution that are usable. [Maximum Size: 500].
Assay name	Assay Name in ASCII string. This matches the test code used in the LIS test order. [Maximum Length: 10]. NOTE: The Atellica Solution only reports assays that are part of the Atellica Solution test map.
Usable Test Count	Calculation of usable tests for the assay. To be usable, the Atellica Solution must enable, calibrate, and QC the test and have consumables to perform these tests. The Atellica Solution reports test counts per assay, not per reagent. The test count of an assay is dependent on the reagent with the lowest level. Typically, the Atellica Solution calculates the test count as follows: The available reagent volume and the required reagent volume is equal to the assay test count.

Reagent inventory response message	Notes
Assay status	0x01 – Green: Usable Test Count of Assay $\geq$ Threshold. 0x02 – Yellow: Usable Test Count of Assay $<$ Threshold. 0x03 – Red: Usable Test Count of Assay = 0. The Threshold value between Green and Yellow is configurable per assay on the Atellica Solution (Default: 10, Range: 1–999).

The LAS can retry the Test Inventory Request if the system does not receive the response after a configurable timeout period. The default value for the Test Inventory Response Timeout is 20 seconds, with a range from 2–60 seconds.

About Onboard Sample Info Messages

The Onboard Sample Info message synchronizes knowledge of the list of sample tubes that the LAS delivers to the Atellica Solution. The message includes a list of sample IDs that the LAS loads the Atellica Solution and a list of sample IDs the LAS loads and removes from the Atellica Solution, but does not unload to the track.

During connection initialization, the LAS sends an Onboard Sample Info Query to synchronize the onboard sample tube information between the LAS and the Atellica Solution.

If the Atellica Solution loses custody of a delivered tube because the operator removes it or a system malfunction occurs, the Atellica Solution sends an unsolicited Onboard Sample Info Response.

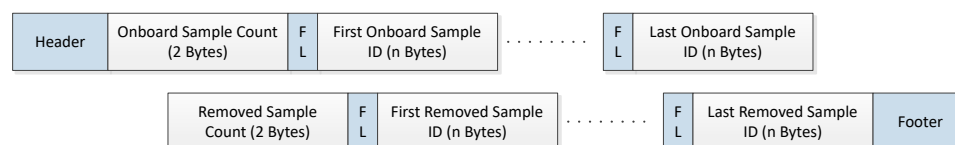
NOTE: If the Atellica Solution loses custody of a delivered tube, the LAS must remove the removed sample tube from its database so that if the operator re-introduces the sample tube to LAS, the Atellica Solution does not declare it as a duplicate sample.

Header	Footer
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Onboard sample info request message**Notes**

Message direction	LAS to Atellica Solution
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Message type	0x0207
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**Onboard sample info response message****Notes**

Message direction	Atellica Solution to LAS
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Usage	Message is sent upon request or is sent unsolicited to indicate a sample removal.
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Message type	0x208
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Onboard Sample Count	Number of sample tubes that the LAS has loaded on the Atellica Solution. [Maximum Count: 90].
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Onboard Sample ID	Zero or more Onboard Sample IDs in ASCII string. [Maximum Length: 24]. NOTE: This sample ID is same sample ID the Atellica Solution uses during sample loading. The Atellica Solution accepts sample ID up to length 20.
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Removed Sample Count	Number of sample tubes that the LAS has loaded and removed since the last update and are not returned to the track. [Maximum Count: 90].
----------------------	---

Onboard sample info response message	Notes
Removed Sample ID	Zero or more Removed Sample IDs in ASCII string. [Maximum Length: 24]. NOTE: This sample ID is same sample ID the Atellica Solution uses during sample loading. The Atellica Solution accepts sample ID up to length 20.

The LAS can retry the Onboard Sample Request if the LAS does not receive the response from the Atellica Solution after a configurable timeout period. The default value for the Onboard Sample Request is 20 seconds, with a range from 2–60 seconds.

About Consumable Inventory Messages

The Consumable Inventory provides the operator and the LAS with additional statuses on Atellica Solution readiness to conduct specific assays. When the LAS requests the consumable inventory, the Atellica Solution responds with inventory for all active consumables. When the Atellica Solution sends an unsolicited consumable inventory response message, it only sends only the consumables that have a changed value since last transmission. After resetting communications, the Atellica Solution performs as if it did not send any previous reagent inventory. The system sends a complete inventory after a communications restore. For example, during Connection Initialization, the LAS sends a Consumable Inventory Request message and the Atellica Solution sends a complete Consumable Inventory. The LAS can request complete consumable inventory only during connection initialization. The Atellica Solution periodically sends unsolicited updates that contain only those consumables with a changed status.

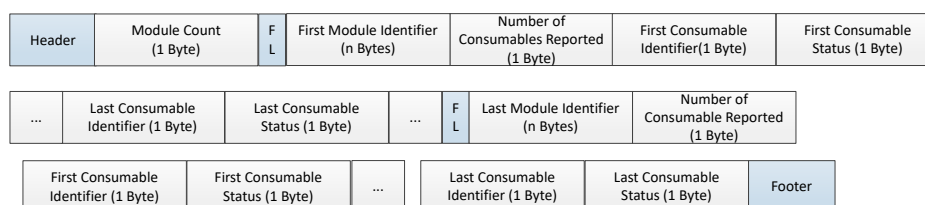
The response message contains only immediately usable consumables based on the status.

Consumable Inventory Request Message:

Header	Footer
--------	--------

Message info	Notes
Message direction	LAS to Atellica Solution
Message type	0x020B

Consumable Inventory Response Message:



Message Info	Notes
Message Direction	Atellica Solution to LAS.
Usage	- The system sends a message upon request. - Instrument replies with consumable status for all consumables. - LAS has a view on the overall consumable data for a particular module using this message. The message facilitates the individual consumable status.
Message Type	0x020C
Module Count	Number of modules or instruments connected to the Atellica Solution. Maximum number of supported analyzers in the deployment is 7 as per the Atellica architecture.
Module Identifier	Analytical module's serial number. Maximum number of characters in the serial number is 15.

Message Info	Notes
Consumable Reported count	Number of consumables with its name and status. Maximum number of consumables per module is 18.
Consumable Identifier	0x01 – CH Cleaner 0x02 – CH Conditioner 0x03 – CH Wash 0x04 – CH Diluent 0x05 – Pretreatment 0x11 – IMT Std B + salt Bridge 0x12 – IMT Standard A 0x13 – IMT Diluent 0x14 – A- LYTE Multisensor 0x21 – IM Acid 0x22 – IM Base 0x23 – IM Cleaner 0x24 – IM Wash 0x25 – Tips 0x26 – Cuvettes 0x27 – Water 0x28 – Cuvette and Tips waste 0x29 – Tip Tray waste
Consumable Status	0x01 – Green: Consumables are in an acceptable state. 0x02 – Yellow: Consumable level low or consumable expires soon (for example, within the next hour). 0x03 – Red: Consumable expired or consumable depleted.

The LAS can retry the Consumable Inventory Request if the response from the analyzer is not received after a configurable timeout period of 2–6-seconds (default 20).

About Initialization Sequence Completed Messages

The Initialization Sequence Completed message notifies the LAS that the 6 messages in (Page 87 *About Initializing the LAS Connection*) are complete. The LAS initiates the sample/carrier routing requests when it receives this message. This is an unsolicited message without any request. After sending Acknowledgement of the analyzer's Handshake message, the LAS starts a timer indicating the start of the Initialization Sequence.

Initialization Completed Message:

Header	Footer
--------	--------

Message info	Details
Message direction	Atellica Solution to LAS
Message type	0x020D

The LAS attempts the reset connection procedure if this message is not received from the analyzer after a configurable timeout period:

Configuration Items	Default Values
InitializationSequenceComplete-WaitTimePeriod	30 seconds (range: 10–600 seconds)

About LAS Connection Reset

When a communication failure occurs, a connection reset must occur. The following are causes for a communication failure:

- Communication Driver issues
- Application software issues
- Hardware disconnect
- Power loss

The LAS protocol contains multiple mechanics to support a connection reset.

When the connection is lost, the Atellica Solution resets Queue knowledge except for locked positions, and the LAS resets Instrument Health, Test Inventory, Consumable Inventory Information, and Transfer Status. While the LAS is disconnected, no pallets move in any queue.

If communication was lost while the Atellica Solution locks the interface position via a Load_Unload Request Command then the LAS does not move any carriers until it receives a Load_Unload Response with the Automation Interface Status of not Critical.

The LAS or the Atellica Solution detects communication failure when timeouts for ACK/NACKs or expected Response Message timeouts are exceeded.

When either side (LAS or Atellica Solution) detects that the connection is not functioning, then each side is responsible for resetting its connection status. After restoring the connection, both sides wait prior to initiating reconnect attempts (Page 82 *About Establishing LAS Connection*).

Wait period timeout configuration items	Default values
Instrument Initial Wait Period (Prior to Entering Listening Mode)	15 second (range 10–600)
Automation Initial Wait Period (Prior to initiating Handshake)	30 second (range 10–600)

About ACK/NACK Timeouts

All messages the LAS or Atellica Solution send must be acknowledged. If the system does not receive an ACK or NACK within the specified timeout, it must re-send the message. After re-sending the message the maximum number of retries without an ACK or NACK response, the sender should assume that the connection is offline and must be reset.

NOTE: The following configuration items' value match between LAS and the Atellica Solution:

ACK/NACK configuration items	Default values
ACK/NACK timeout (from LAS)	1 second (range 1–9)
ACK/NACK timeout (from the Atellica Solution)	1 second (range 1–9)
Maximum Allowed Retries	5 (range 1–9)

About Response Message Timeouts

When the Atellica Solution sends an acknowledged request message or command message, a corresponding response message always occurs. The operator can send a message on Acknowledgment timeout, but the system cannot retrieve it if the response times out. The following is a summary table of the response timeouts:

Category	Item	Range of values	Default	Units (data type)
Instrument Health	Response Timeout	2–60	20	seconds (int)
Test Inventory	Response Timeout	2–60	20	seconds (int)
Onboard Sample Info	Response Timeout	2–60	20	seconds (int)
Transfer Status	Response Timeout	2–60	20	seconds (int)
Add Queue	Response Timeout	2–60	20	seconds (int)
Skip Queue	Response Timeout	2–60	20	seconds (int)
Clear Queue	Response Timeout	2–60	20	seconds (int)
Load_Unload Command	Response Timeout	60–900	600	seconds (int)
Consumable Inventory	Response Timeout	2–60	20	seconds (int)

About NACK Responses

When the Atellica Solution sends a message, there always must be an acknowledgment of that message. The Atellica Solution generates a NACK acknowledgment message when the checksum does not match or a received message contains invalid values. If the message acknowledgment is a NACK response, the operator must resend the message. If the Atellica Solution receives the maximum NACK acknowledgment messages for a message, then the operator must reset the connection.

NACK response configuration items	Default values
Maximum NACK limit for a message	3 (range 1–9)

About Keep Alive Messages

During LAS connection inactivity, Keep Alive messages monitor the connection. Keep Alive messages are not request messages, therefore no corresponding response messages exist. When the Atellica Solution receives a Keep Alive message, the receiver only needs to acknowledge via ACK/NACK response.

Header	Footer
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Keep alive	Notes
Message direction	LAS to Atellica Solution
Message type	0x0005
Keep alive message	Default values
Inactivity timeout before sending Keep Alive message	15 (range 5–300) seconds

About Carrier Queue Management

A series of queue messages informs the Atellica Solution of carriers diverted to the LAS buffer. The Atellica Solution uses this information to anticipate and prepare sample loading, unloading, or exchange. By preparing for sample transfer, the Atellica Solution ensures a high rate of exchange.

About Add Queue Command Messages

As the Atellica Solution prepares for a load, unload, or exchange operation, the LAS must send an Add Queue message to the Atellica Solution in the order of divert. The Atellica Solution expects the carrier to be presented for sample transfer in first-in-first-out (FIFO).

For SHC, automation must send Add Queue commands to respective IPs. The Atellica Solution sends an Add Queue command to IP0 when a carrier diverts to IP0 queue. When the Atellica Solution adds a carrier to IP1 via released from IP0 or diverts directly from the main track via an IP0 divert gate, the Atellica Solution must send an Add Queue command for IP1.

NOTE: The Atellica Solution accepts a duplicate Sample ID in an Add Queue message. If the Atellica Solution detects a duplicate Sample ID, it returns the sample with a duplicate sample error.

NOTE: The Atellica Solution sets STAT priority when a sample tube contains STAT order or if it introduces a sample tube via Priority Lane.

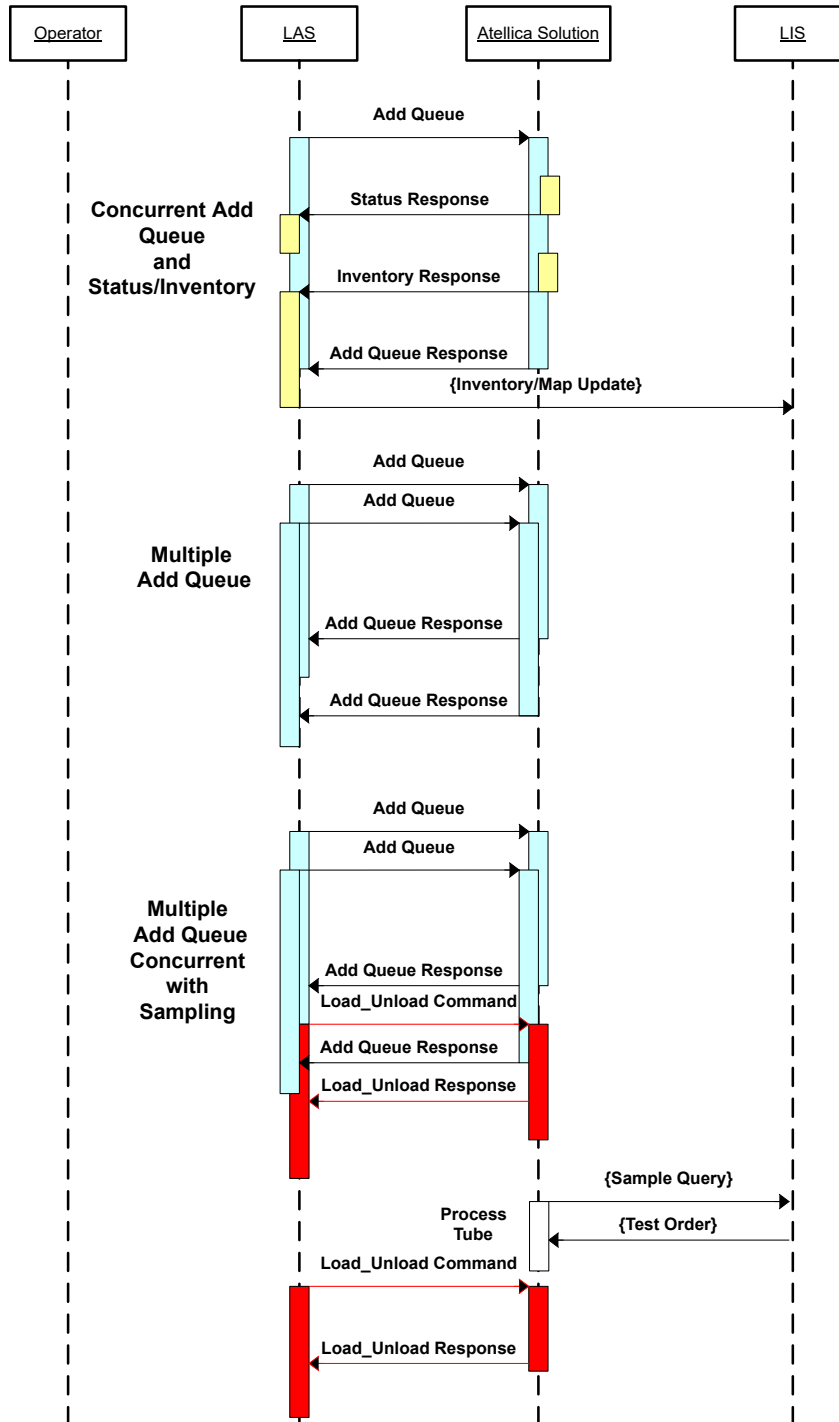
NOTE: The Atellica Solution will accept Greiner MiniCollect Complete samples only when it is configured with Atellica Solution Immunoassay analyzers because the Greiner MiniCollect Complete sample is required to go for volume check before routing to perform the actual test.

Header	Interface Position Index (1 Byte)	Carrier Occupancy (1 Byte)	F L	Sample ID (n Bytes)	Sample Priority (1 Byte)	Tube Height (1 Byte)	Tube Diameter (1 Byte)	Footer
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Add queue command request message	Notes
Message Direction	LAS to Atellica Solution.
Message Type	0x0401.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.
Carrier Occupancy	0x01: Empty Carrier. 0x02: Uncapped Tube. 0x03: Capped Tube. NOTE: If the carrier occupancy is empty, a sample ID does not exist (Field Length for Sample ID must be set to 0). If carrier occupancy type is Capped Tube or Uncapped Tube, then a sample ID must exist.
Sample ID	Sample ID in ASCII string [Maximum Length: 24].
Sample Priority	Priority of Sample tube in the carrier. 0x00 – Undefined. 0x01 – Routine. 0x02 – STAT.
Tube Height	The Atellica Solution ignores it (Unit: Millimeter).

Add queue command request message	Notes						
Tube Diameter	Unit: Tenths of a millimeter 0x52 – Inside diameter is 8.2 mm. The tube is in the category of Greiner MiniCollect Complete tubes. 0x82 – the tube is in the category of 13 mm outside diameter tubes 0x96 – the tube is in the category of 15 mm outside diameter tubes 0xA0 – the tube is in the category of 16 mm outside diameter tubes Value 0x52 identifies Greiner MiniCollect Complete tubes Tubes with Tube Diameter of 0x52 will be considered as Greiner MiniCollect Complete tubes by Atellica Solution. Any other diameter value from the above list or outside of the list will be processed as a regular (i.e. not Greiner MiniCollect Complete) tube by Atellica Solution.						
<table><tr><td>Header</td><td>Interface Position Index (1 Byte)</td><td>F L</td><td>Sample ID (n Bytes)</td><td>Command Status (1 Byte)</td><td>Footer</td></tr></table>		Header	Interface Position Index (1 Byte)	F L	Sample ID (n Bytes)	Command Status (1 Byte)	Footer
Header	Interface Position Index (1 Byte)	F L	Sample ID (n Bytes)	Command Status (1 Byte)	Footer		
Add queue command response message	Notes						
Message Direction	Atellica Solution to LAS						
Usage	The Atellica Solution sends a message upon request. LAS cannot initiate a Load_Unload Command until it receives an Add Queue Response. The Atellica Solution can send more than 1 Add Queue Command at a time.						
Message Type	0x0402.						

Add queue command response message	Notes
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.
Sample ID	Sample ID in ASCII string [Maximum Length: 24].
Command Status	0x01 – The Atellica Solution performs Add Queue Command successfully. NOTE: The system cannot reject the Add Queue command. However, the system is not required to load or process the sample.



About Skip Queue Command Messages

When a carrier has been added to the queue, the Atellica Solution can issue the Skip Queue Command at any time before the carrier is locked by a Load_Unload Command request. Upon receipt of Skip Queue Command, the Atellica Solution removes the skipped carrier from its carrier queue database. When there is an ambiguity, such as multiple samples with duplicate sample IDs, the first item the Atellica Solution adds with matching criteria is skipped. Skip Queue Command affects the matching Sample ID & Carrier Occupancy Type in FIFO order.

The LAS must wait for a Skip Response message and acknowledge response message prior to moving the pallet out of IPO/IP1 gate.

NOTE: The Atellica Solution can send a Skip Command to any unlocked queue element while another pallet is locked for transfer.

Header	Interface Position Index (1 Byte)	Carrier Occupancy (1 Byte)	F L	Sample ID (n Bytes)	In Queue (1 Byte)
				Tube Height (1 Byte)	Tube Diameter (1 Byte)
					Footer

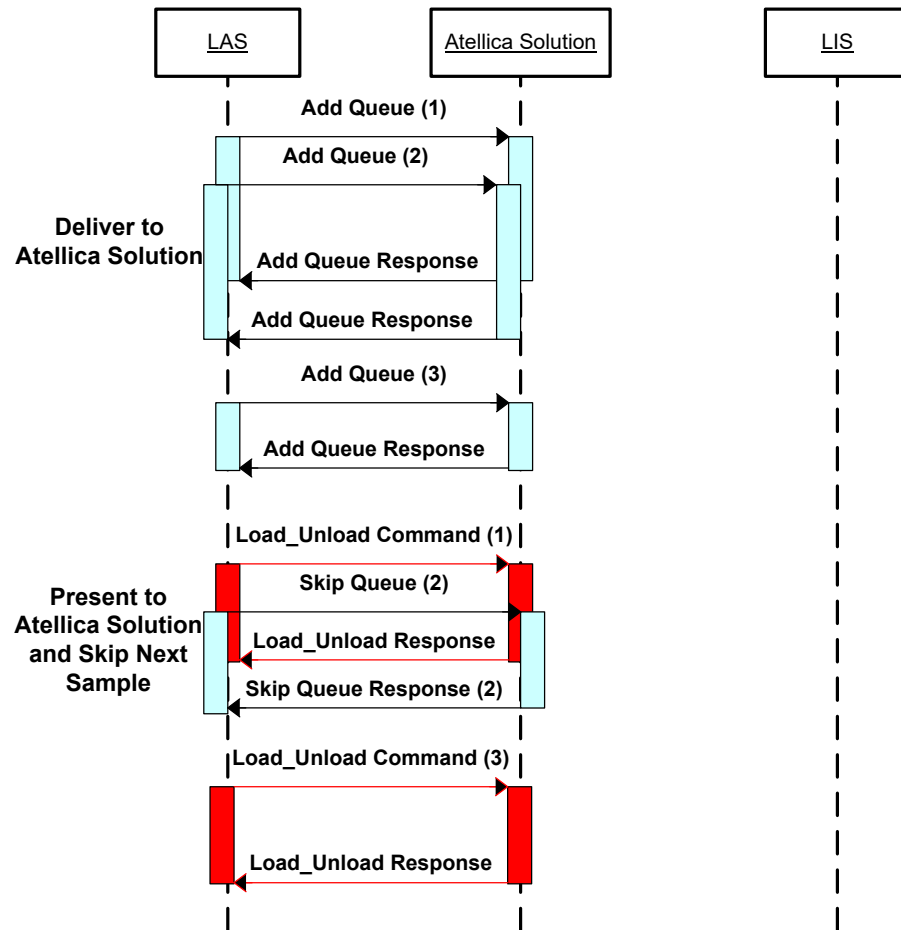
Skip queue command request message	Notes
Message Direction	LAS to Atellica Solution.
Message Type	0x0403.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.
Carrier Occupancy	0x01: Empty Carrier. 0x02: Uncapped Tube. 0x03: Capped Tube. NOTE: If carrier occupancy is empty, a sample ID does not exist. (Field Length for Sample ID must be set to 0). If carrier occupancy type is Capped Tube or Uncapped Tube, then a sample ID must be present.
Sample ID	Sample ID in ASCII string [Maximum Length: 24].

Skip queue command request message	Notes
In Queue	0x00: The Atellica Solution previously added to queue and it is physically on LAS buffer (the typical scenario). 0x01: The Atellica Solution previously added to queue, but it is not physically on LAS buffer (a divert error occurred).
Tube Height	The Atellica Solution ignores it (Unit: Millimeter).
Tube Diameter	The Atellica Solution ignores it (Unit: Millimeter).

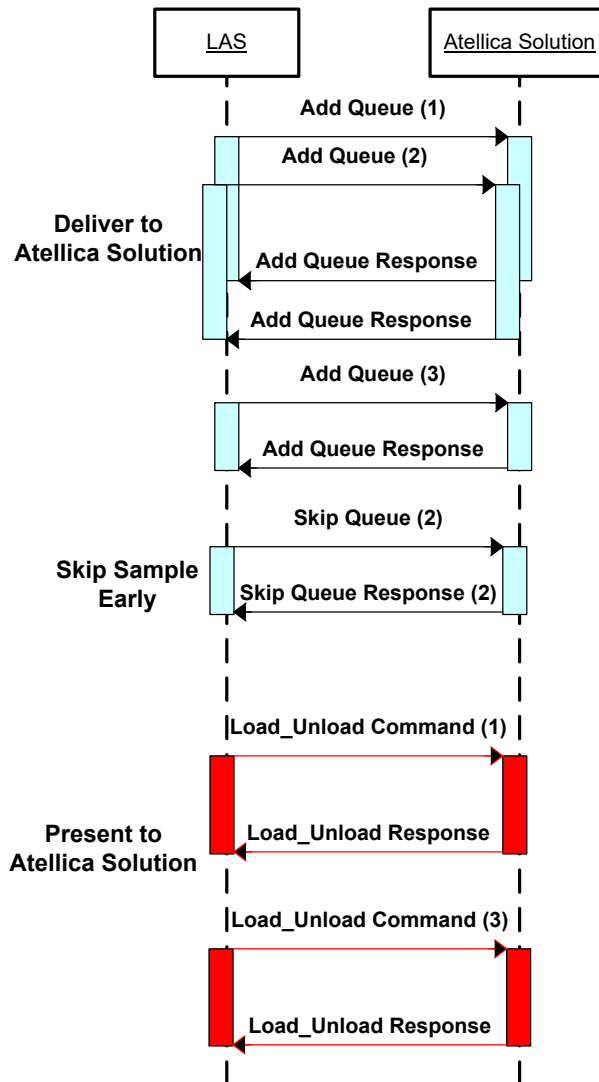
Header	Interface Position Index (1 Byte)	F L	Sample ID (n Bytes)	Command Status (1 Byte)	Footer
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Skip queue command response message	Notes
Message Direction	Atellica Solution to LAS.
Usage	The Atellica Solution sends a message upon request. The Atellica Solution removes the tube from its queue and abort any scheduled processing for this tube if possible.
Message Type	0x0404.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.
Sample ID	Sample ID in ASCII string. [Maximum Length: 24].
Command Status	0x01: The Atellica Solution performs Skip Queue Command successfully.

Typical message with skip command sequence diagram



Multi-cycles early skip command sequence diagram



Clear Queue Command Message

This command informs the Atellica Solution to clear all carriers presently in the buffer from its database. While automation can send a Clear Queue Command at any time, it does not affect the carrier already the Load_Unload Command locks.

When the Atellica Solution receives a Clear Queue Command Response, the cleared carriers can move through the Interface Position without requiring the Load_Unload Command message exchanges. If a locked carrier is in the Interface Position, then the clear carrier waits until the locked carrier becomes unlocked.

Header	Interface Position Index (1 Byte)	Footer
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Clear queue command request message details

Notes

Message Direction	LAS to Atellica Solution.
Message Type	0x0405.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer. For the SHC, the Atellica Solution must send separate Clear Queue commands for IPO and IP1 during communication initialization. However, it is not always necessary to clear both queues as a pair during run time if only 1 queue needs clearing.

Header	Interface Position Index (1 Byte)	Command Status (1 Byte)	Footer
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Clear queue command response message

Notes

Message Direction	Atellica Solution to LAS.
Usage	The Atellica Solution sends a message upon request. The Atellica Solution removes all queued tubes it has previously added but not sampled. The Atellica Solution must send tubes queried and not sampled with null test results and an aborted error flag to the LIS. The Atellica Solution should process any result tube that was successfully sampled normally.

Clear queue command response message	Notes
Message Type	0x0406.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.
Command Status	0x01: The Atellica Solution performs Clear Queue Command successfully.

Instrument Control

About Transfer Status Messages

The Transfer Status message informs LAS of immediate readiness of the Atellica Solution to Load or unload tubes. The LAS uses this information to initiate Load_Unload Command and empty pallet routing decisions. The LAS does not base sample routing decisions on Transfer Status as the Atellica Solution does not generate necessary consumables if no samples are waiting to load. Instead, LAS uses Analyzer Health and Test Inventory messages to route samples.

Header	Interface Position Index (1 Byte)	Footer
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Message info	Notes
Message Direction	LAS to Atellica Solution.
Message Type	0x0209.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.

Header	Interface Position Index (1 Byte)	Ready To Load (1 Byte)	Return Ready Tube Count (2 Bytes)	Footer
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Transfer status response message	Notes
Message Direction	Atellica Solution to LAS.
Message Type	0x020A.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.
Ready to Load	Message that indicates that the Atellica Solution is ready to Load. 0x00: Not Ready to Load. 0x01: Ready to Load.
Return Ready Tube Count	The number of tubes immediately ready to return to LAS.
Transfer status message configuration item	Default values
Transfer Status Response Timeout	20 seconds (range 2–60)

About Load_Unload Command Messages

This command supports sample pick-and-place by the Atellica Solution. The Load_Unload Command message indicates to the Atellica Solution that a carrier is locked for the Atellica Solution to perform sample transfer operations. When the Atellica Solution Transfer Arm is clear of the track, the Atellica Solution transmits the Load_Unload Command response to the LAS. At this point, the LAS can release the carrier from its locked position and move the next carrier into the lock position. While the carrier is locked, the Atellica Solution expects to have full access to the locked carrier space. The track cannot unlock the carrier until the Atellica Solution receives the response message. If the LAS releases a sample prematurely, a mechanical jam, damage, or sample spill can occur.

NOTE: The LAS must wait for Load_Unload Command response to unlock the position and release the carrier. If the LAS receives a Health Status message while waiting for a Load_Unload Command response, the LAS ignores Lock Ownership status. The LAS can release the carrier upon receipt of the Load_Unload Command Message response only if neither the Load Command Status nor Unload Command Status are 0x02 – Error performing Load command: Lock Carrier in place or Error performing Unload command: Lock Carrier in place.

If the operator performs a communication reset is performed internally or externally and the Instrument Health Response Message indicates that the lock ownership status is Not Locked by the instrument, the LAS can release the carrier without waiting for a Load_Unload Command Message response.

After mixing a whole blood sample, the LAS sends the sample to the Atellica Solution. The maximum time for whole blood samples to be loaded onto the Atellica Solution from the LAS is 8 minutes. If a sample is in the Load Queue and the maximum time is reached, the LAS skips the sample. The operator can configure this element in the LAS.

Configuration items	Default values
Analyzer Shaking Timeout	8 minutes (range: 2–30 minutes)

NOTE: For the SHC, all solicited Load_Unload Command response message advances queue by 1 in the Atellica Solution's database.

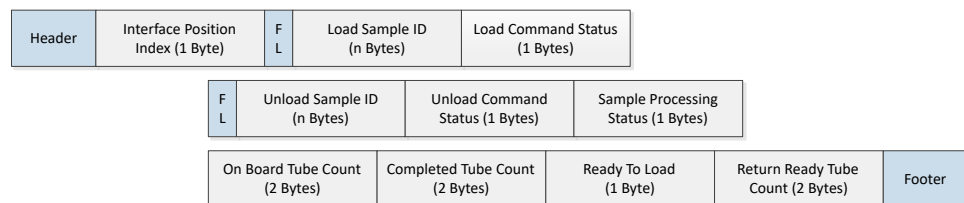
NOTE: The Atellica Solution checks if the maximum needed volume for ordered tests exceeds a certain limit (100 uL). If the needed volume exceeds the limit then the Atellica Solution rejects the orders and returns the sample tube without processing, and sample processing status code 0x22 – Insufficient Sample is utilized.

Greiner MiniCollect Complete tubes are processed similar to capillary tubes. Every time a tube is sent to CH analyzers, it is first sent to IA to check for minimum volume availability. If there are no connected IM analyzers, then Greiner MiniCollect Complete tubes are not processed. Sample processing status code, 0x15 Not Processed Capability Unavailable, is used.

Header	Interface Position Index (1 Byte)	Carrier Occupancy (1 Byte)	F L	Sample ID (n Bytes)			
					Tube Height (1 Byte)	Tube Diameter (1 Byte)	Footer

Load_Unload Command request message	Notes
Message Direction	LAS to Atellica Solution.
Message Type	0x0303.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.
Carrier Occupancy	0x01: Empty Carrier. 0x02: Uncapped Tube. 0x03: Capped Tube. NOTE: If carrier occupancy is empty, a sample ID does not exist (Field Length for Sample ID must be set to 0). If carrier occupancy type is Capped Tube or Uncapped Tube, then a sample ID must be present.
Sample ID	Sample ID in ASCII string. [Maximum Length: 24]
Tube Height	The Atellica Solution ignores it (Unit: Millimeter). Unit: Tenths of a millimeter 0x52 – Inside diameter is 8.2 mm. The tube is in the category of Greiner MiniCollect Complete tubes. 0x82 – the tube is in the category of 13 mm outside diameter tubes 0x96 – the tube is in the category of 15 mm outside diameter tubes 0xA0 – the tube is in the category of 16 mm outside diameter tubes Value 0x52 identifies Greiner MiniCollect Complete tubes Tubes with Tube Diameter of 0x52 will be considered as Greiner MiniCollect Complete tubes by Atellica Solution. Any other diameter value from the above list or outside of the list will be processed as a regular (i.e. not Greiner MiniCollect Complete) tube by Atellica Solution.
Tube Diameter	The Atellica Solution ignores it (Unit: Millimeter).

Load_Unload Command request message	Notes
Elapsed Time	The elapsed time in seconds from the LAS initiating the Load Unload Request to the Atellica Solution accepting the whole blood sample. Elapsed time starts when the system releases the sample from the Shaker Module. The default value for this parameter is 0xFFFFh.



Load_Unload Command response message	Notes
Message Direction	Atellica Solution to LAS.
Usage	The Atellica Solution sends a message upon request. The Atellica Solution replies with Sample ID, Command Status, On Board Tube count, and Completed Tube count.
Message Type	0x0304.
Interface Position Index	The 0-based index value of Interface Position. Index Position starts closest to sample entry position (divert gate) to the LAS buffer.
Load Sample ID	Sample ID in ASCII string [Maximum Length: 24]. Field length maximum is 24, but the Atellica Solution does not accept sample IDs greater than 20 characters. When the Atellica Solution loads a sample from the LAS, the Sample ID returns in ASCII. This does not mean that the Atellica Solution verifies the Sample ID with barcode reader. The Atellica Solution performs Sample ID verification after sample transfer and before processing the sample. If a transfer error occurs without disturbing the loaded sample tube, such as an arm jam during lateral move from the Atellica Solution side to the LAS side, then the field matches LAS provided value.

Load_Unload Command response message	Notes
Load Command Status	The Load Command Status reports the transfer status of sample 0x01: The Atellica Solution performs Load Command successfully. - The Atellica Solution successfully loaded the original sample in pallet. 0x02: The Atellica Solution error performing Load command: Lock Carrier in place. - Transfer Arm failure occurs within the LAS space. Automation Interface Status is Critical (Page 89 <i>Instrument Health Message</i>). The LAS must not release the carrier and informs the operator. - The Atellica Solution cannot move the pallet until the Instrument Health Status for Automation Interface Status changes to Green or Red. - The Atellica Solution successfully accesses pallet content. - Pallet is empty when Instrument Health Status for Automation Interface Status changes to Green or Red. Accessed sample tubes are considered manually removed from the LAS and the Atellica Solution. - Sample load detection generates an Onboard Sample Info response. Sample loss may vary by physical condition. 0x03: Error performing Load command: OK to Unlock Carrier. - Transfer Arm failure occurs but is not interfering with the track so the carrier can be released. Automation Interface Status is Red. - Pallet is full and it is moveable. - The Atellica Solution has not accessed pallet content. - Transferring sample tubes are considered manually removed from the LAS and the Atellica Solution. - Sample load detection generates an Onboard Sample Info response. Sample loss may vary by physical condition.

Load_Unload Command response message	Notes
	0x04: Unable to perform command: Queue Mismatch. - When carriers do not match the Atellica Solution queue knowledge, this value is returned. - Pallet is not affected and the LAS can deliver it to another analyzer. NOTE: The Atellica Solution responds with this status when a carrier, which is not in the queue or is out of sequence, is requires transfer. 0x05: Unable to perform command: Interface position is offline. - The Atellica Solution is in the offline or local control mode and does not load any further samples until it becomes online mode. - Pallet is not affected and the LAS can deliver it to another Atellica Solution. NOTE: To be consistent, the Atellica Solution assumes the unlocked carrier is released from carrier queue. Optionally, if the LAS needs to hold carrier in position to retry at a later time, then it may clear queue and then rebuild queue via Add Queue commands without flushing the buffer (this is the LAS' prerogative) including the supposedly released carrier in front. 0x06: Load Skipped. - The Atellica Solution may return this status when it presents a loaded carrier to an Unload Only position. - The Atellica Solution returns this status if the interface position's Remote Control Mode is Unload Only. - Pallet is not affected and the LAS can deliver it to another analyzer. 0x07: Unable to perform command: Instrument Skipped Loading.

Load_Unload Command response message	Notes
	• The Atellica Solution skips loading sample for an internal issue: – Automation Interface Status change. – Instrument Process Status change. – LIS Connection Status change. – Complete Test Inventory loss. – The Atellica Solution rejects loading on capped tube. – Pallet is not affected and the LAS can deliver it to another analyzer. 0x08: Unable to perform command: Unsupported Sample ID. • Either the Sample ID exceeds supported length or uses unsupported characters [Supported SID Maximum length: 20 characters].
Unload Sample ID	• Sample ID in ASCII string [Maximum Length: 24]. • If an empty carrier in lock position exists, the Atellica Solution may unload LAS sample to LAS. The operator can lock or generate either empty carrier by loading the LAS sample into the Atellica Solution. If the Atellica Solution did not load a tube, the field is left blank. • This is the same Sample ID the LAS provided to the Atellica Solution when it loaded the sample from the LAS to the Atellica Solution.

Load_Unload Command response message	Notes
Unload Command Status	0x01: The Atellica Solution Performs Unload Command successfully. - Pallet is full and is movable. 0x02: Error performing Unload command: Lock Carrier in place. - An error occurs such that the Transfer Arm is blocking the track and the carrier cannot be released. The Atellica Solution sets the Automation Interface Status to Critical (Page 89 <i>Instrument Health Message</i>). The LAS informs the operator. - Pallet is not movable until Instrument Health Status for Automation Interface Status changes to Green or Red. - The Atellica Solution accesses pallet content. - Pallet is empty when Instrument Health Status for Automation Interface Status changes to Green or Red. - Accessed sample tube is considered manually removed from the LAS and the Atellica Solution. - The Atellica Solution generates an Onboard Sample Info response upon sample loss detection. Sample loss may vary by physical condition. 0x03: Error performing Unload command: OK to Unlock Carrier. - An error occurs but the Transfer Arm is not interfering with the LAS and the carrier can be released. Automation Interface Status is Red. - Pallet is empty and is movable. - The Atellica Solution has not access pallet content. - Transferring sample tube is considered manually removed from the LAS and the Atellica Solution. - Sample loss detection generates an Onboard Sample Info response. Sample loss may vary by physical condition. 0x04: Unable to perform command: Queue Mismatch.

**Load_Unload Command
response message**

Notes

- When carriers do not match the Atellica Solution queue knowledge, this value is returned.
- Pallet is not affected and the Atellica Solution can deliver it to another analyzer.
- **NOTE:** The Atellica Solution responds with this status when a carrier, which is not in the queue or is out of sequence, requires transfer.

0x05: Unable to perform command: Interface position is offline.

The Atellica Solution is in the offline or local control mode and does not unload samples until it becomes online mode.

Pallet is not affected and the LAS can deliver it to another analyzer.

NOTE: The Atellica Solution assumes the unlocked carrier is released from carrier queue. If the LAS needs to hold carrier in position to retry at a later time, it can clear queue and rebuild queue via Add Queue commands without flushing the buffer, including the supposedly released carrier in front.

0x06: Unload Skipped.

- The Atellica Solution may return this status when it presents an empty carrier to a Load Only position.
- The Atellica Solution returns this status if the interface position's Remote Control Mode is Load Only.
- Pallet is not affected and the LAS can deliver it to another analyzer.

0x07: Unable to perform command: Instrument Skipped Unloading.

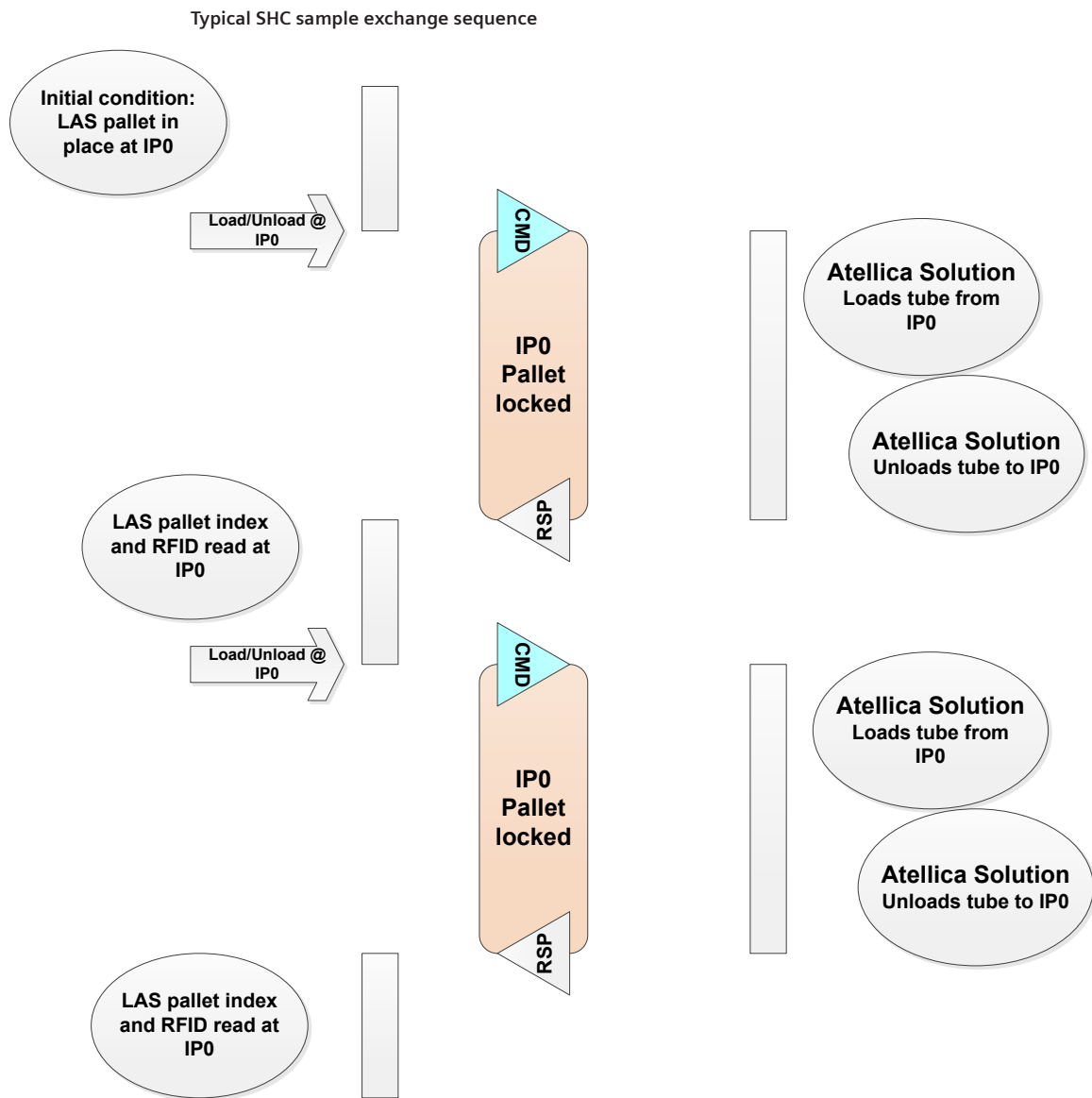
The Atellica Solution skipped unloading sample for internal issue:

Automation Interface Status changes.

Pallet is not affected and the LAS can deliver it to another Atellica Solution.

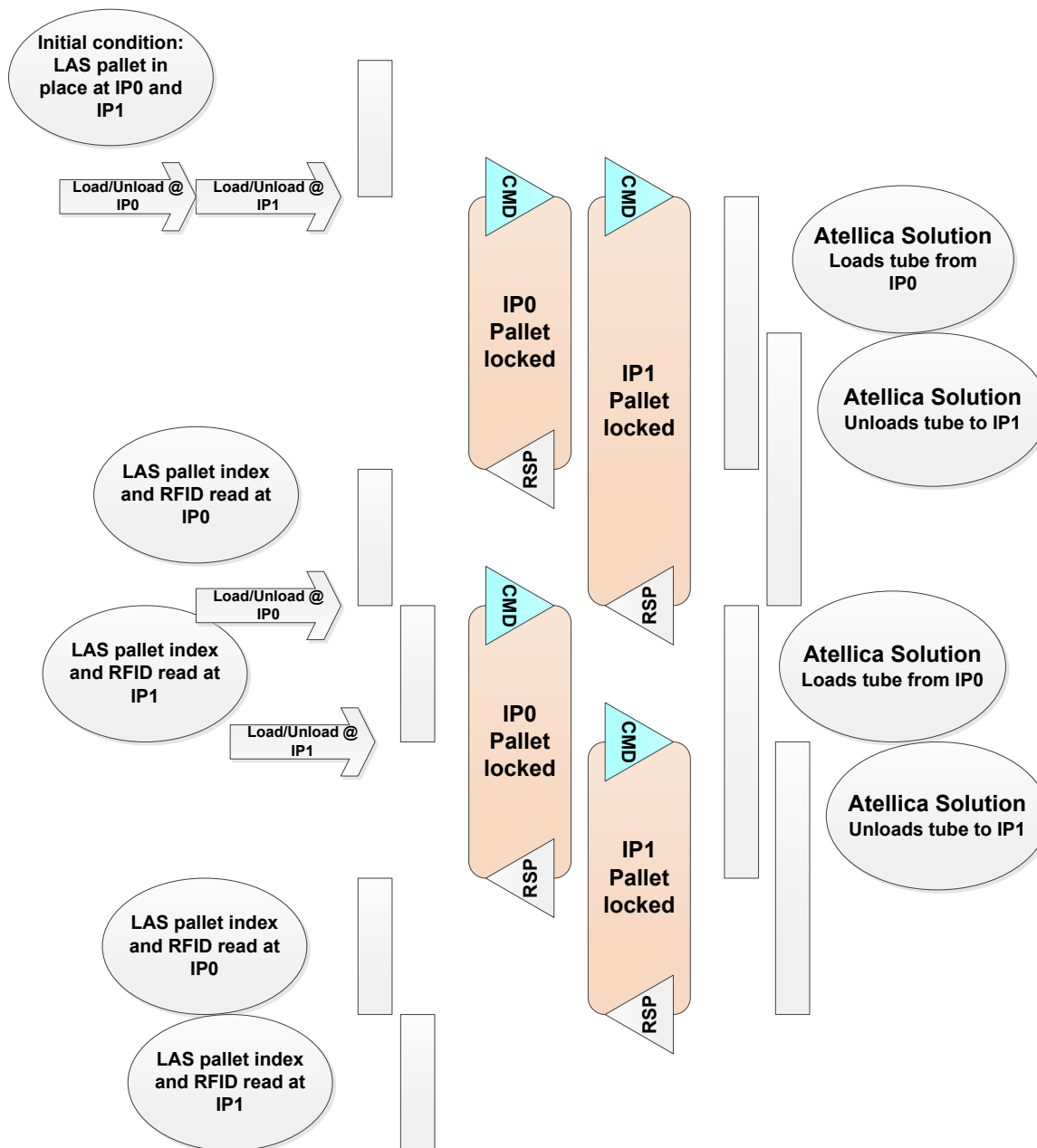
Load_Unload Command response message	Notes
Sample Processing Status	Sample Processing Status provides 2 types of information: • Order Processing Status: – Completed Order processing – Order Processing Attempted – Order processing NOT Attempted • Sample Condition: – Sample Usable – Sample Error (Operator Inspection Required) 0x00: No Tube Unloaded. 0x01: Sample Processed successfully. 0x11: Sample not processed: Instrumentation failure. • For example, a Sample Tube Identifier Error. 0x12: Sample not processed: Sample ID mismatch. • After sample loading, the Atellica Solution tries to verify the Sample ID and finds a mismatch between the provided barcode and read barcode. 0x13: Sample not processed: Unreadable Barcode. • After sample loading, the Atellica Solution fails to read Sample ID. 0x14: Sample not processed: No LIS orders. • The Atellica Solution finds no orders for the unloaded sample to process. 0x15: Sample not processed: No consumables/Not Processed Capability Unavailable. • The Atellica Solution is out of consumables and needs to process an unloaded sample. • The Atellica Solution is not available to perform the volumetric check. 0x16: Sample not processed: No reagent. • The Atellica Solution is out of reagents and needs to process an unloaded sample. 0x17: Sample not processed: Duplicate Sample Error.

Load_Unload Command response message	Notes
	0x18: Sample not processed: Capped Tube. 0x22: Error processing sample: Insufficient sample. - The Atellica Solution detects insufficient samples. 0x23: Error processing sample: Clogged sample. 0x26: Error processing sample: Unknown Container Type. 0x27: Error Processing sample: HIL Problem / Specimen Type Issue. - The sample tube's orders contain whole blood samples and the Atellica Solution does not support whole blood or flags the sample with HIL. 0x28: Error Processing sample: WholeBloodMixingToSamplingTimeExpired - The sample tube orders contain a whole blood sample with an expiration date occurring prior to aspiration of all the ordered tests.
On Board Tube Count	Number of tubes onboard from LAS.
Completed Tube Count	Number of tube onboard ready to be return to LAS.
Ready to Load	Indicates that the Atellica Solution is ready to Load. 0x00: Not Ready to Load. 0x01: Ready to Load. (Page 120 <i>About Transfer Status Messages</i>)
Return Ready Tube Count	The number of tubes immediately ready to return to LAS. (Page 120 <i>About Transfer Status Messages</i>)



NOTE: The Atellica Solution independently sends Load_Unload Command requests to each Interface Position.

Load_Unload command requesting sent independently to each interface position



About SHC IP1 Unlock Delayed

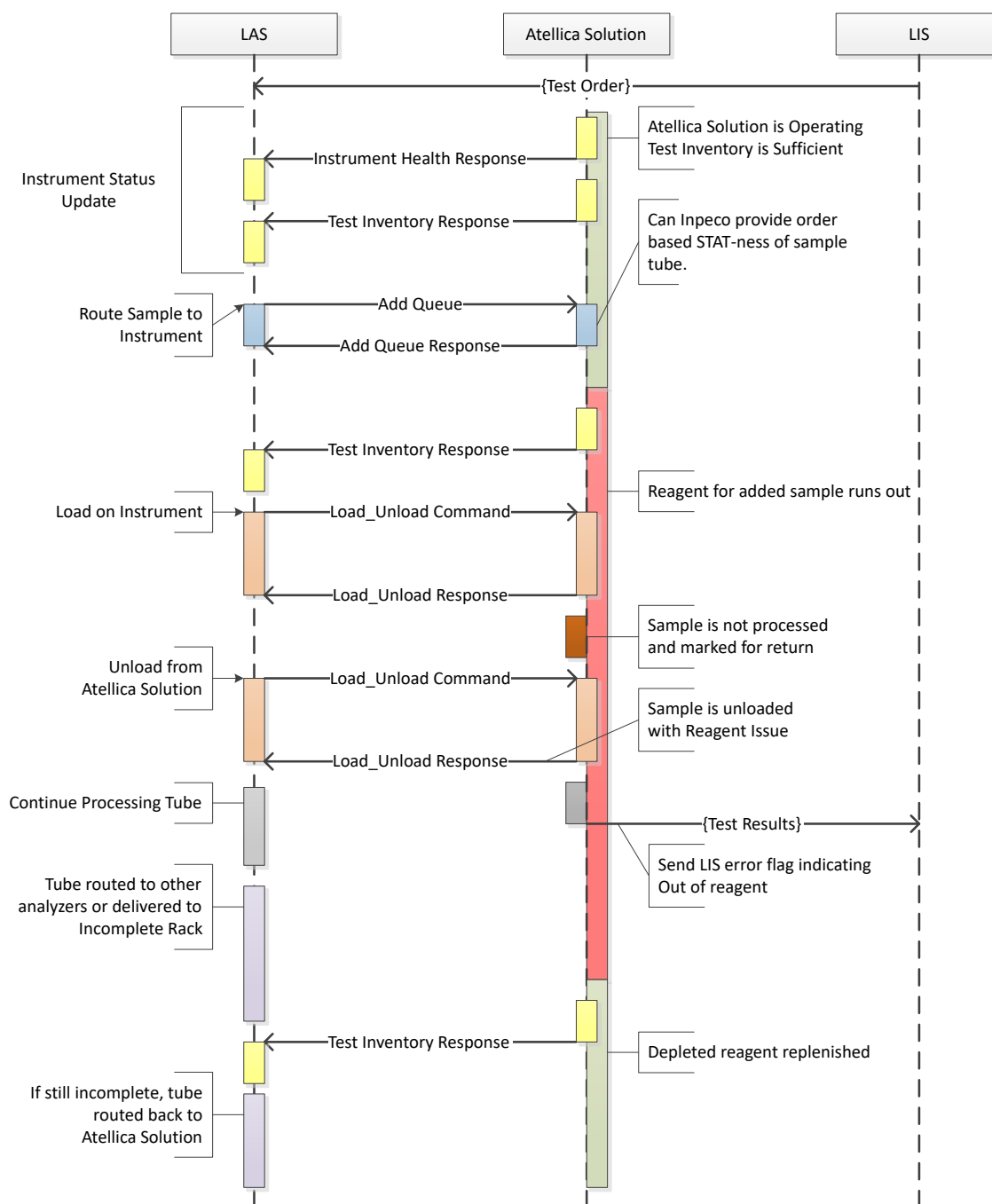
Full carriers diverted to the Atellica Solution enter the IP0 queue. When a load operation releases carriers from IP0, they move to the IP1 queue for an unload operation.

These queues can hold only a finite number of carriers. If the Atellica Solution delays releasing carriers from IP1, the IP1 queue can fill up which delays any operations at IP0. This may also cause the IP0 queue to fill up which may delay additional carriers being diverted to the Atellica Solution.

The LAS skips carriers in IP1 to generate space in IP1 queue.

About Out of Reagent Scenario

At times, reagent may be out for samples that the operator loads onto the instrument. These samples return with processing status and a reagent issue flag (Page 100 *About Test Inventory Messages*) (Page 104 *About Consumable Inventory Messages*).



About Instrument Status Change Scenario

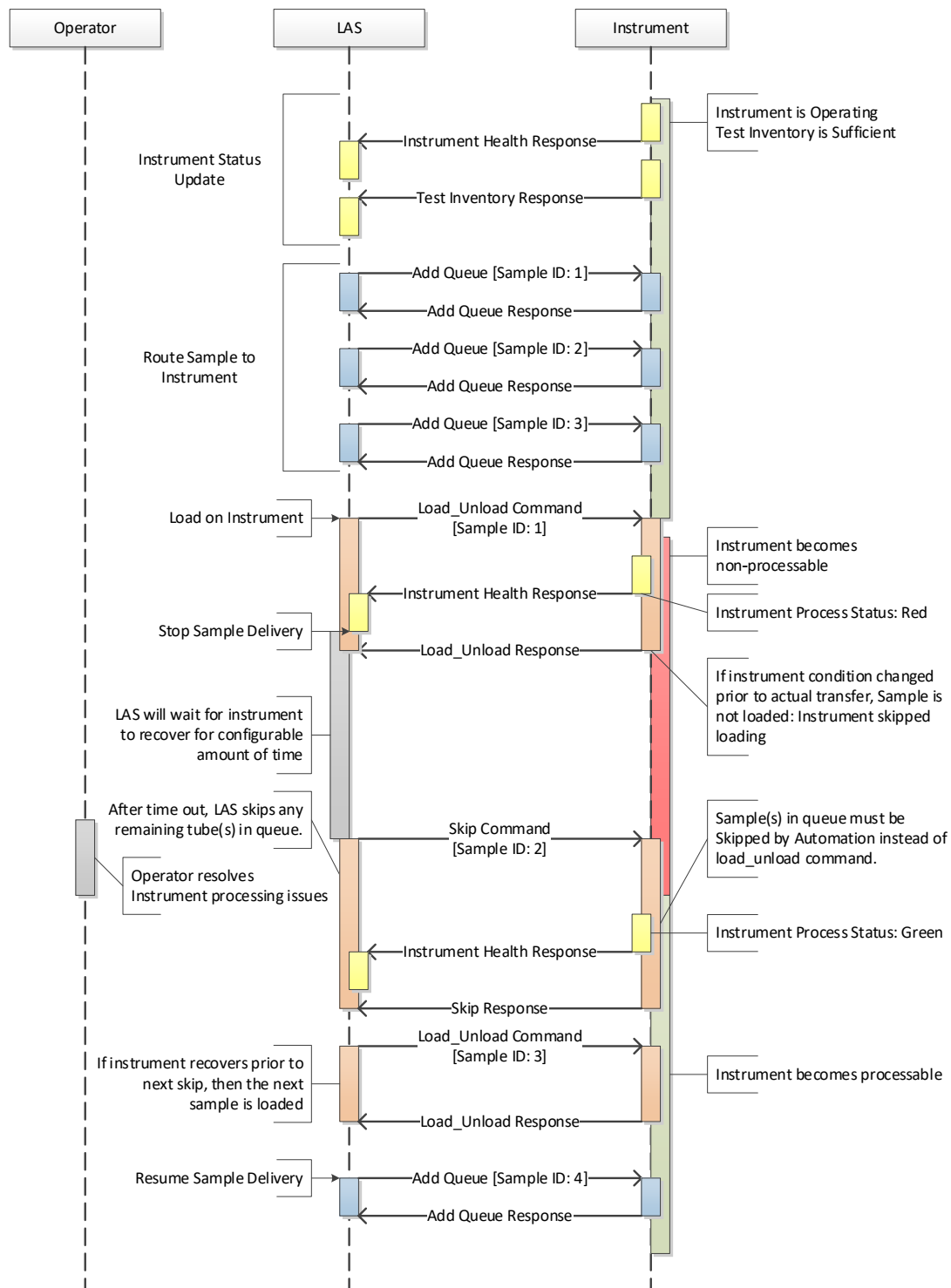
The Instrument Health Status may change so that sample is no longer loadable. If the Atellica Solution receives a Load_Unload Command request before reporting the Instrument Health Status, the Atellica Solution sends an unsolicited Instrument Health response followed by a Load_Unload Command response indicating that sample was not loaded (Page 87 *About Initializing the LAS Connection*).

Sample Load/Processing Affecting Status [Any]

Status	Type	Notes
Automation Interface Status	All	0x03: Red 0x04: Critical
Remote Control Status	SHC	- Offline: - IP0: 0x01: Offline - IP1: 0x01: Offline - Unloading Only: - IP0: 0x01: Offline - IP1: 0x05: On-Line Unloading Only Mode
LIS Connection Status	All	0x02: Disconnected
Instrument Process Status	All	0x03: Red

Sample Unload Affecting Status [Any]

Status	Type	Notes
Automation Interface Status	All	0x03: Red 0x04 – Critical
Remote Control Status	SHC	- Offline: - IP0: 0x01: Offline - IP1: 0x01: Offline



6

Algorithm

Set of rules or processes for solving a specific problem.

ASTM

ASTM International formerly known as American Society for Testing and Materials.

Atellica Data Manager

Siemens middleware product connecting the laboratory LIS and analyzers/automation systems in the laboratory. In addition, it provides a variety of test request and test result data processing and data analysis tools to the laboratory. Abbreviation: Atellica DM.

Atellica Sample Handler

An Atellica Solution sample management module with 4 drawers for loading samples, a Cal/QC tube storage area, and a robot that moves samples between the loading drawers, the storage area, and the Atellica Magline Transport. Abbreviation: SH.

Automatic transfers

Automatic results reporting and automatic work-list entry requests.

Buffer

Data storage to compensate for a difference in the flow rate of information or the time that events occur when transmitting data from 1 device to another.

Checksum

A variable, 2 digits, hexadecimal number included in data link message frames to support error detection.

Communication parameters

Characteristics of a communication connection that the Atellica Solution configures.

Control character

Character that provides information about transmitted data or to control data transmission.

Data link layer

Layer 2 of the ASTM communication protocol. Ensures reliable transmission of data across the physical medium.

Data link message

Logical unit of data that contains application data segments along with additional data that provides transmission synchronization and error protocol information.

Flow control

Any method that regulates incoming data in an effort to have the time and resources to process the incoming data without losing any of it.

Frame of reference

An orthogonal co-ordinate system such that X is horizontal and parallel to the direction of sample carrier motion on Automation track, Y is horizontal and perpendicular to X, and Z is vertical.

Half duplex

A method that describes data link duplexity where both stations may transmit, but only 1 at a time.

Handshaking

Mechanism for coordinating the transmission of data across a channel.

Interface

Interfaces are the functional, physical, and informational characteristics that exist at common boundaries between items (modules, sub-systems, components). Interfaces allow systems, equipment, software, personnel and data to be made compatible with each other.

Interface position

A location in an LAS buffer where sample containers are positioned for interaction with equipment connected to the LAS. Equipment includes analyzers connected to the LAS as well as LAS modules performing various pre-analytical and post-analytical sample processing. Different LAS manufacturers use different terms for the Interface Position: pick position, place position, aspirate position, stop gate, sampling gate, and so on. Abbreviation: IP.

Interface specification

The allowable values for all the characteristics of an interface between components to ensure those connected components achieve the desired higher level functionality. In addition, an interface specification defines requirements for the interfacing components to ensure the desired higher level functionality is achieved when those components are connected.

LAS Buffer

An LAS component where sample containers are presented for processing by a connected analyzer or other module. Interface Positions are part of an LAS buffer.

LIS ASTM compliant

LIS systems can use various communication protocols in order to send and receive data with the system. An ASTM compliant LIS system conforms to the ASTM standards for data communication.

LIS HL7 compliant

LIS systems can use various communication protocols in order to send and receive data with the system. An HL7 compliant LIS system conforms to the HL7 standards for data communication.

Load (sample)

The action of transferring a sample container from an LAS to the Atellica Solution.

Message

Data string with a specific meaning transmitted over a communication channel. It always has a descriptive word preceding it, for example, an "application" message.

Message transfer phase

Second phase of an ASTM data link session, application message data is transmitting during this phase.

Middleware

Software product connecting between the LIS and the analyzers and automation system that consolidates test orders and results.

Operator

The user of an Atellica Solution in a laboratory performing ordered tests for physicians.

Pallet

An LAS component that carries sample containers on the LAS. Different LAS manufacturers use different terms: puck, pallet, sample carrier, and so on.

Parity checking

Character error detection code that uses the binary digits of the data bits and any existing parity bit, in which the total number of 1s in each encoded character is even or odd.

Physical layer

Layer 1 of the ASTM communication protocol, concerned with the electrical, mechanical and timing aspects of the signal transmission over a medium.

Pick and place sampling

The LAS presents a sample container at the LAS buffer interface position. The connected analyzer removes the sample container from the LAS for processing. The connected analyzer returns the sample container to the LAS after processing.

Planner

An Atellica Solution component that coordinates activities across the various module types within the analyzer. It sends action requests to the Sample Source and Sinks, such as Load or Unload a sample, and so on. Within the command, it also passes sample information such as priority and sorting group and is responsible for routing the sample tubes containers to the specific Analytical Modules for processing.

Point in Space Sampling

The automation system presents an uncapped sample container at the track sampling position. The connected analyzer moves a sample aspiration probe to the sample container on the track and aspirates sample for the requested tests.

Protocol

Set of conventions that apply to the format and relative timing of the way messages transmit between the communicating devices.

R-232

Example of a physical layer standard for an interconnection of computer systems.

Remote system

Device on the other end of the physical connection between 2 communicating devices. An example of a remote system is the LIS, a general purpose computer or test equipment that is connected to an analyzer.

Sample, Source, and Sinks

Subsystems where samples are added, removed, or stored for the Atellica Solution.

Sample Handler Connect

A sample transfer unit that connects an Atellica Solution to a Lab Automation System. The Sample Handler Connect uses 2 robots to transfer routine and STAT samples to and from the Lab Automation System. Abbreviation: SHC.

System analyzer manager

The central unifying component for the Atellica system. It is responsible for coordinating information across each Atellica module, including the SH, maintaining the sample inventory, and managing test orders and actions among the various parts of the system. The SIM sends commands to the SHC to switch state, query status and execute actions. Abbreviation: SIM.

System of systems

A collection of task-oriented or dedicated systems that pool their resources and capabilities together to create a new, more complex system that offers more functionality and performance than simply the sum of the constituent systems.

Unload (sample)

The action of transferring a sample container from the Atellica Solution to an LAS pallet.

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